

Twelve Days Between Observations: Validation-First Flood Risk Assessment and Emergency Routing for a Single-Orbit Coastal District

Abstract

Over Dakshina Kannada, Karnataka, exactly one Sentinel-1 orbit covers the district on a strict twelve-day cycle, so no individual flood there can be mapped from synthetic aperture radar (SAR): a flash flood rises and recedes inside the revisit gap. We treat that constraint as the design premise, decomposing risk into a static susceptibility surface $S(\mathbf{x})$ trained on a multi-temporal SAR flood-*frequency* inventory and a dynamic rainfall trigger $T(t)$, fused as $R(\mathbf{x}, t) = S(\mathbf{x}) \cdot T(t)$. A remote-sensing confound is identified and removed: a model given land cover and NDVI reaches ROC-AUC 0.9878, but SHAP shows it has learned that closed canopy hides water from SAR, not flood physics. The terrain-and-rainfall model scores 0.9625 on a random split, 0.9474 under spatial-block cross-validation, and 0.871 transferring to a held-out city window never seen in training, where its highest-susceptibility tenth contains 49.5% of all SAR-observed flooding. We show the calibrated score is tied to the balanced training prior and must not be read as a flood probability without correction. The trigger, keyed to India Meteorological Department rainfall categories and fitted to no event, ranks two documented floods above ordinary and dry conditions but separates a flood from a heavy-rain non-flood day by only 0.011, a margin governed almost entirely by its antecedent-saturation term. Over a 28,529-edge road graph (17.1% flood-prone), flood-aware routing cuts mean exposure by 33% for a 3.8-minute detour, holding at 31% over 200 randomly sampled incident sites; a survivorship-corrected 7×7 sweep shows the outcome is insensitive to the penalty weight but sits beside a topological cliff in the impassability threshold. A* matches Dijkstra’s optimal path on all 42 hospital-incident pairs while exploring a median 31.1% fewer nodes, yet runs faster on only 25. A 9-qubit QUBO/QAOA dispatch layer reproduces the Hungarian optimum exactly, with no advantage claimed at this scale.

Keywords: flood susceptibility; SAR flood inventory; spatial cross-validation; explainable AI; flood-aware routing; emergency dispatch

1 Introduction

Flooding is the most frequently recurring natural hazard in India, and the country’s south-western coastal belt concentrates several of the mechanisms that make it difficult to forecast at a useful spatial scale. Dakshina Kannada district, centred on the port city of Mangaluru in coastal Karnataka, receives more than 4000 mm of rainfall in an average monsoon, drains two substantial rivers, the Nethravathi and the Gurupura, into a shared estuary, and does so across a coastal plain narrow enough that fluvial discharge and tidal stage interact within a few kilometres of the built-up area. Extreme-rainfall frequency over this stretch of the Western Ghats and the Karnataka coast has been shown to be trending upward [1], while land-use change in the Netravathi basin has altered its runoff response [2, 3] and the district’s land cover has itself changed measurably over recent decades [4], so the hazard is neither stationary nor purely meteorological. Two recent events frame this study: the August 2024 Nethravathi river flood and the May 2025 Mangaluru urban flood. Existing flood work on this basin is predominantly static and index-based, for

example the GIS and analytic-hierarchy susceptibility zoning of Nirmala [5], which produces a fixed map with no temporal component.

The dominant response in the literature is flood-susceptibility mapping: train a classifier on terrain and hydrological conditioning factors against an inventory of observed flooding, and produce a wall-to-wall map of relative propensity [6–9]. The approach is mature, and the recent generation pairs it with explainable-AI attribution so that the resulting map can be interrogated, not merely displayed [10–12]. Our work sits in this tradition, and departs from it in three specific ways that this paper’s contributions address.

The observational premise is usually left implicit. Susceptibility studies typically source their inventory from whatever flood observations are available, then treat the resulting map as if it were a spatial model of flooding in general. But the density of the underlying satellite record constrains what such a map can honestly claim. Over Dakshina Kannada, Sentinel-1 [13] coverage consists of a *single* orbit geometry, a descending pass on relative orbit 63, giving a nominal revisit of approximately twelve days. This matters because rapid SAR inundation mapping of the kind demonstrated elsewhere in India [14] presumes an acquisition close enough in time to the peak to be representative. We verified this directly against the Copernicus archive instead of assuming the constellation’s global six-day figure applies. The consequence is strict: no individual flood in this district can be mapped from SAR, because the hydrograph of a coastal flash flood peaks and recedes well inside the revisit gap. Work that frames flood mapping as per-event spatial classification implicitly assumes a denser record than exists here. Section 4.1 sets out the two-stage decomposition that follows from taking this seriously.

Validation is usually optimistic. Random k -fold cross-validation on geospatial samples leaks information across fold boundaries, because neighbouring points share terrain; the resulting score measures interpolation within sampled neighbourhoods, not skill at an unvisited location [15–17]. Reported AUCs above 0.95 are common in this literature and are rarely accompanied by a spatial control. We report random k -fold, spatial-block k -fold, and a strict geographic-transfer test in which the model is trained entirely outside the evaluation window (Section 5.3).

Free parameters are usually asserted. Flood-aware routing formulations introduce cost-function constants, a penalty weight and an impassability threshold, that are typically fixed by judgement and never swept. Because these constants determine which roads the router will refuse to use, they are exactly the parameters an operational deployment would be questioned on. Section 5.6 reports the full 7×7 sweep and finds a result that is more nuanced than a stability claim.

1.1 Contributions

1. A two-stage risk decomposition, $R(\mathbf{x}, t) = S(\mathbf{x}) \cdot T(t)$, in which a SAR-flood-frequency-trained static surface is modulated by an independently validated live trigger. The decomposition is derived from a measured satellite-revisit constraint rather than adopted for convenience.
2. Identification and ablation of a remote-sensing circularity that inflates susceptibility scores in forested coastal catchments, where SAR cannot observe water under canopy and a classifier given land cover learns detectability rather than hazard.
3. A three-level validation ladder, random split, spatial-block cross-validation, and geographic transfer, quantifying how much reported skill survives each successive removal of spatial leakage.

4. A complete sensitivity analysis of the routing cost function that distinguishes a genuinely insensitive parameter from one adjacent to a topological discontinuity.
5. A search-transparent Dijkstra/A* benchmark across 42 hospital-incident pairs, reporting a wall-clock result that runs counter to the node-count result and explaining why.
6. A hybrid QUBO/QAOA dispatch layer verified against the exact classical optimum, scoped explicitly as a feasibility demonstration.

Figure 2 maps the full pipeline.

2 Related Work

2.1 Machine learning for flood susceptibility

Tree ensembles dominate the field. Tehrany et al. [6] established the SVM-and-frequency-ratio baseline; Khosravi et al. [7] benchmarked multi-criteria against machine-learning approaches; Arabameri et al. [8] and Saha et al. [9] established random forest and boosted-tree variants as the practical default. Reported AUCs cluster between 0.85 and 0.95. Wang et al. [18] and Avand et al. [19] extended the approach to data-scarce basins. The conditioning factors are largely standardised: elevation, slope, curvature, the topographic wetness index [20], height above nearest drainage [21, 22], distance to channel, drainage density and rainfall climatology.

Explainability entered the field recently. Lundberg and Lee [10] introduced SHAP; Aydin and Iban [11] and Seydi et al. [12] applied it to flood susceptibility, generally to confirm that elevation and channel proximity dominate. Our SHAP analysis is used differently, as a diagnostic that exposed a confound (Section 4.5), not as a post-hoc confirmation.

2.2 Spatial validation

That geospatial models are routinely over-validated is established but under-applied. Roberts et al. [15] formalised block cross-validation for spatially structured data; Ploton et al. [16] demonstrated that ignoring it inflates reported accuracy in remote-sensing regressions; Meyer and Pebesma [17] extended the argument to area-of-applicability estimation. Flood-susceptibility papers reporting spatial CV remain a minority, which is part of the motivation for Section 5.3.

2.3 SAR flood mapping

SAR change detection for inundation is well established [23–25], typically with backscatter thresholding against a dry reference and a permanent-water mask [26]. The recognised limitation is that C-band cannot detect water beneath closed canopy [27, 28], which is precisely the mechanism behind the confound we isolate. Revisit-limited inventories are usually handled by aggregating across many passes into a frequency product [26, 29], the approach adopted here. Closest to the present work, Sadhwani and Eldho [30] combine SAR with elevation data in a dual machine-learning scheme for inundation mapping *and* depth estimation over the Indian west-coast belt, using the same sensor and terrain inputs as this study. Their depth-regression stage is precisely the extension our Limitation 1 identifies as the highest-value next step, and the contrast is instructive: they target an absolute physical quantity, while the index reported here is explicitly relative.

2.4 Emergency routing and dispatch

Risk-weighted shortest-path routing under flooding has an established literature [31, 32]. A* [33] remains the standard goal-directed refinement of Dijkstra [34], with admissibility guaranteeing

optimality. Emergency-vehicle siting and allocation is itself a mature operations-research field, from maximal-covering formulations [35] to the ambulance location and relocation models surveyed by Brotcorne et al. [36]; our contribution is not a new formulation but the coupling of that assignment to a SAR-grounded, time-varying risk surface. Ambulance-to-incident assignment is a classical linear assignment problem solved exactly by the Hungarian method [37], which is why any quantum treatment at this scale must be framed as feasibility, never as advantage.

2.5 Quantum optimization

QAOA [38] approximates combinatorial optima on gate-model hardware; Lucas [39] gives the standard Ising encodings, including assignment. Zhou et al. [40] introduced the INTERP parameter-transfer heuristic used here. Known obstacles, barren plateaus [41] and noise-limited depth, mean that near-term demonstrations at practical scale remain out of reach; our scoping in Section 5.8 reflects that.

2.6 Gap

No prior work we are aware of combines an inventory that is explicit about single-orbit revisit limits, a trigger validated against documented events without fitting, a routing layer with a complete parameter sweep and a search-transparent algorithm comparison, and a dispatch layer verified against the exact classical optimum, applied to one Indian coastal district with every number reproducible from public data.

3 Study Area and Data

3.1 Study area

Dakshina Kannada district occupies the coastal plain and lower Western Ghats scarp of southwestern Karnataka (Figure 1). The modelling domain is the bounding box $[74.65^\circ, 12.65^\circ]$ to $[75.40^\circ, 13.15^\circ]$; detailed routing and validation are conducted over the Mangaluru city window $[74.78^\circ, 12.80^\circ]$ to $[74.92^\circ, 13.00^\circ]$. Table 1 summarises the physical setting as measured from the assembled data stack, not quoted from secondary sources.

Three features of the setting drive the methodology. First, relief is extreme over short distances: elevation in the sampled floodable domain spans -5 m to 279 m, with the Ghats escarpment rising steeply inland, so runoff concentration times are short and flood peaks are flashy. Second, the coastal strip is low and tidally influenced, so the estuary can impound river discharge, a coupling that a rainfall-only trigger captures only indirectly. Third, land cover in the low-lying non-flooded fraction of the district is predominantly closed-canopy vegetation, which is what makes the SAR detectability confound of Section 4.5 bite.

3.2 Documented events

Two events anchor the temporal validation. The August 2024 Nethravathi flood followed sustained catchment wetting, 216.9 mm over the preceding 72 h, with a comparatively modest 53.4 mm in the final 24 h. The May 2025 Mangaluru urban flood combined a high 24-hour total (87.2 mm, IMD “Heavy”) with a similarly saturated antecedent state (205.6 mm/72 h). Their contrasting structure, one saturation-driven, one intensity-driven, is what makes them a useful pair for testing a trigger that weights both. Full inputs are in Table 9.

Table 1: Study-area characteristics. Terrain and rainfall statistics are computed over the sampled floodable-lowland domain ($n = 3000$ points); network statistics are from the OpenStreetMap drivable graph of the Mangaluru window.

Property	Value	Source
District	Dakshina Kannada, Karnataka, India	FAO GAUL
Modelling bounding box	74.65°–75.40°E, 12.65°–13.15°N	–
Routing window	74.78°–74.92°E, 12.80°–13.00°N	–
Principal rivers	Nethravathi, Gurupura (shared estuary)	MERIT Hydro
Elevation range (sampled)	–5 m to 279 m	SRTM
HAND range (sampled)	0 m to 79.4 m	MERIT Hydro
Mean annual rainfall	3373 mm to 4401 mm (mean 4124)	CHIRPS 2020–2024
Monsoon regime	South-west, June–September	IMD
Sentinel-1 coverage	1 orbit (descending, rel. orbit 63)	Copernicus
Nominal SAR revisit	12 d	Copernicus
Road graph	11,910 nodes, 28,529 edges	OpenStreetMap
Flood-prone edges ($S > 0.5$)	4873 (17.1%)	this study
Hospitals modelled	7	OpenStreetMap
Incident sites modelled	7	this study

3.3 Data sources

All inputs are public and ingested programmatically (Table 2). Satellite and terrain layers are processed server-side on Google Earth Engine [42]; only sampled and aggregated outputs are downloaded.

4 Methodology

4.1 Why the risk model is decomposed

A fully spatiotemporal flood model would estimate inundation $I(\mathbf{x}, t)$ jointly in space and time. Fitting such a model requires observations resolved at the timescale of the process. With one usable SAR orbit at a twelve-day cadence, the observational record cannot constrain the temporal dimension at all. We therefore factor the problem:

$$R(\mathbf{x}, t) = S(\mathbf{x}) \cdot T(t), \quad S(\mathbf{x}) \in [0, 1], \quad T(t) \in [0, 1], \quad (1)$$

where $S(\mathbf{x})$ is a time-invariant relative propensity estimated from the *frequency* of observed flooding across all available passes, and $T(t)$ is a spatially uniform scalar estimated from meteorological data available at hourly cadence. The factorisation assigns each term to the data that can actually constrain it. Its cost is an explicit modelling assumption, that the spatial pattern of relative risk does not change with event magnitude, which we return to in Section 5.10.

4.2 SAR flood-frequency inventory

For each monsoon window (1 June–30 September 2024; 15 May–30 September 2025), every Sentinel-1 GRD scene matching the fixed orbit geometry is speckle-filtered with a 30 m focal median and thresholded for open water:

$$W_k(\mathbf{x}) = \mathbb{K}[\sigma_{VV,k}^0(\mathbf{x}) < -16 \text{ dB}] \cdot \mathbb{K}[\sigma_{VH,k}^0(\mathbf{x}) < -22 \text{ dB}], \quad (2)$$

where $\sigma_{VV,k}^0$ and $\sigma_{VH,k}^0$ are calibrated backscatter coefficients for pass k . A pixel counts as *flooded* on pass k only if it is water then and dry in a January–February baseline, and is not permanent

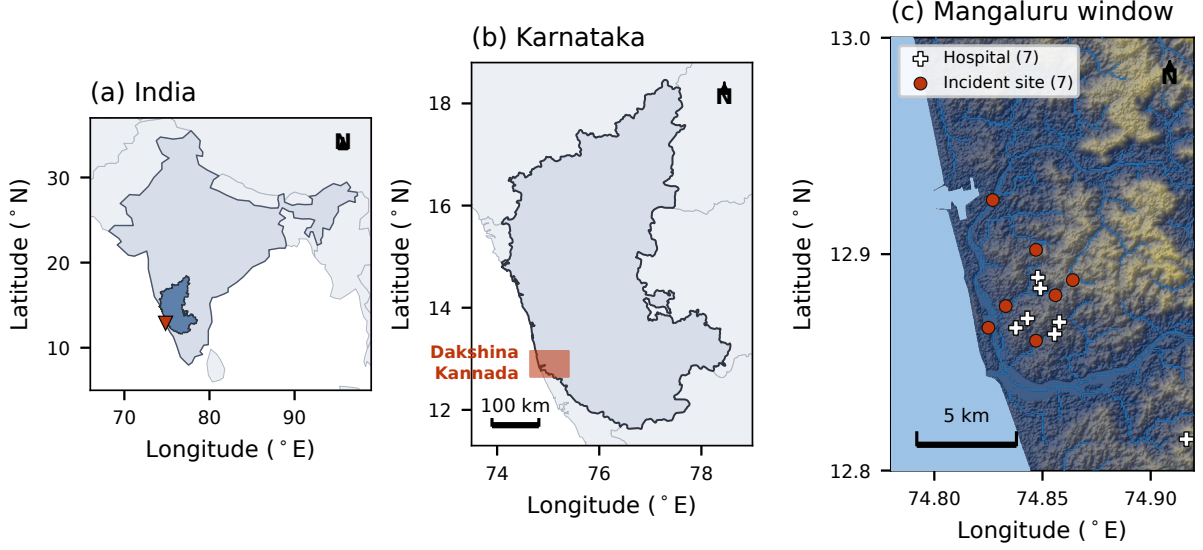


Figure 1: Study area. (a) India with Karnataka shaded and Mangaluru marked. (b) Karnataka with the Dakshina Kannada modelling box. (c) The Mangaluru routing window: hillshaded SRTM terrain, the MERIT-derived channel network, the sea and estuary, and the seven hospitals and seven incident sites used throughout. The coastline is the valid-data boundary of the SRTM stack, not a cartographic overlay.

water:

$$F_k(\mathbf{x}) = W_k(\mathbf{x}) \cdot (1 - W_{\text{dry}}(\mathbf{x})) \cdot \mathbb{I}[O_{\text{JRC}}(\mathbf{x}) < 80\%], \quad (3)$$

with O_{JRC} the JRC Global Surface Water occurrence [26]. The inventory is the normalised frequency over the $K = 20$ qualifying passes:

$$\Phi(\mathbf{x}) = \frac{1}{K} \sum_{k=1}^K F_k(\mathbf{x}) \in [0, 1]. \quad (4)$$

Vectorising $\Phi > 0$ over the routing window yields 201 polygons totalling 2.86 km² (Figure 3). This area is a *lower bound* on flooding that occurred, not a flood map: it records only what was inundated at the instants the satellite happened to look.

4.3 Conditioning factors and collinearity screening

Twelve candidate factors were computed (Figure 4). TWI and the stream power index follow the standard definitions [20]:

$$\text{TWI} = \ln\left(\frac{a}{\tan \beta}\right), \quad \text{SPI} = \ln(a \tan \beta), \quad (5)$$

with a the specific catchment area (m²) and β the local slope. HAND is the elevation of a pixel above the nearest hydrologically connected drainage cell [21, 22].

Collinearity was screened by variance inflation factor,

$$\text{VIF}_j = \frac{1}{1 - R_j^2}, \quad (6)$$

where R_j^2 is the coefficient of determination of factor j regressed on all others. As Table 3 shows, TWI and SPI are mutually collinear by construction, both being functions of a and $\tan \beta$; with both present they score 16.22 and 14.50. Removing SPI reduces TWI to 1.49 and leaves every retained factor below 3.5, comfortably under the conventional threshold of 10.

Table 2: Data sources. Every layer is public; none is hand-authored or simulated.

Source	Product	Resolution	Role
Sentinel-1 [13]	GRD IW, VV+VH, rel. orbit 63	10 m	Multi-temporal flood-frequency inventory
JRC GSW [26]	Water occurrence	30 m	Permanent-water exclusion mask
SRTM [43]	Digital elevation model	30 m	Elevation, slope, aspect, curvature
MERIT Hydro [44]	HAND, upstream area	90 m	HAND, TWI, distance to river, drainage density
CHIRPS [45]	Daily precipitation	5 km	Mean annual rainfall climatology
ESA WorldCover [46]	Land cover	10 m	Ablation only (excluded from primary model)
Sentinel-2	Surface reflectance	10 m	NDVI; ablation only
Open-Meteo / ERA5 [47]	Hourly precipitation	point	Live and archived antecedent rainfall
WorldTides	Tide height	point	Acquired for the live dashboard; <i>not</i> used in $T(t)$ (§5.10)
OpenStreetMap	Drivable road network	vector	Routing graph [48]
Natural Earth	Administrative outlines	10 m scale	Cartography only

4.4 Sampling with terrain-matched negatives

Training points are drawn by stratified sampling, 1500 per class, under two constraints. Positives require $\Phi(\mathbf{x}) \geq 0.05$; negatives require $\Phi(\mathbf{x}) = 0$ *and* a distance of at least 120 m from any flood pixel, which removes the ambiguous halo around observed flooding. Both classes are confined to a floodable-lowland domain,

$$\mathcal{D} = \{\mathbf{x} : z(\mathbf{x}) \leq 280 \text{ m} \wedge h(\mathbf{x}) \leq 80 \text{ m}\}, \quad (7)$$

with z elevation and h HAND, both thresholds set at the 90th percentile of the flood-pixel distribution. Without $\mathcal{D}$ the negative class would be drawn largely from forested uplands, and the classifier could separate the classes on elevation alone without learning anything flood-specific. This elevation- and HAND-matching of negatives is, in our assessment, the single decision that most affects whether the reported scores mean anything.

4.5 A remote-sensing confound, and its removal

The first model trained on all twelve factors reached a test ROC-AUC of 0.9878. SHAP attribution, however, ranked land cover and NDVI as the two most important predictors, ahead of elevation and HAND (Figure 7c). Inspecting the sample explained why: within the floodable-lowland domain, non-flood pixels are overwhelmingly closed-canopy vegetation. C-band SAR cannot observe standing water beneath a closed canopy [27, 28]. A pixel under forest is therefore labelled “not observed flooded” for a sensing reason, not a hydrological one, and a model given land cover can achieve high accuracy by learning *where SAR can see the ground*, not where water collects.

This is circularity, not signal, and it would have transferred a spurious model into an operational routing system. We therefore define the primary feature set as terrain and rainfall only, nine factors, and retain the twelve-factor model solely as a sensitivity analysis. Section 5.1 quantifies the cost of this decision: 0.9878 to 0.9625 in test ROC-AUC, with the SHAP ranking reverting to elevation, HAND, distance to river, rainfall and TWI, an ordering consistent with the physical literature.

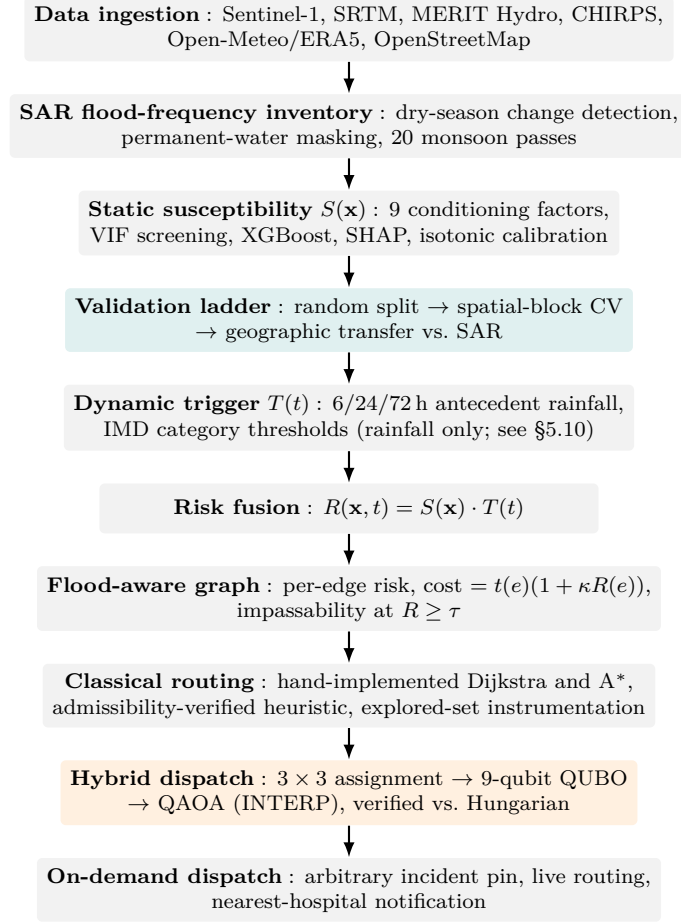


Figure 2: Pipeline. The two-stage split into a static SAR-trained surface and a dynamic live trigger (shaded path through $S(\mathbf{x})$ and $T(t)$) is a direct consequence of the twelve-day single-orbit revisit established in Section 1, not a modelling preference.

4.6 Classifier, calibration and explanation

Random forest [49] and XGBoost [50] were trained on a 70/30 stratified split, with 5-fold cross-validation on the training portion. XGBoost was selected on test ROC-AUC.

Raw tree-ensemble scores are ranking outputs, not probabilities [51], so we calibrate with isotonic regression [52] fitted by cross-validation on the training split alone, and evaluate with the Brier score [53]. Because the training sample is class-balanced while the landscape is not, the calibrated score is tied to a 0.5 prior; Section 5.2 quantifies this and applies the standard prior correction [54, 55]:

$$\text{BS} = \frac{1}{n} \sum_{i=1}^n (\hat{p}_i - y_i)^2, \quad (8)$$

with $\hat{p}_i$ the predicted probability and $y_i \in \{0, 1\}$ the label. Isotonic regression finds the monotone g minimising $\sum_i (g(s_i) - y_i)^2$ over the raw scores s_i , so calibration cannot reorder predictions and therefore cannot alter AUC by construction.

Feature attribution uses SHAP values [10], the Shapley decomposition of a prediction over feature subsets:

$$\phi_j = \sum_{S \subseteq \mathcal{F} \setminus \{j\}} \frac{|\mathcal{S}|! (|\mathcal{F}| - |\mathcal{S}| - 1)!}{|\mathcal{F}|!} [f(\mathcal{S} \cup \{j\}) - f(\mathcal{S})], \quad (9)$$

with $\mathcal{F}$ the feature set; we report $\overline{|\phi_j|}$ over the test split.

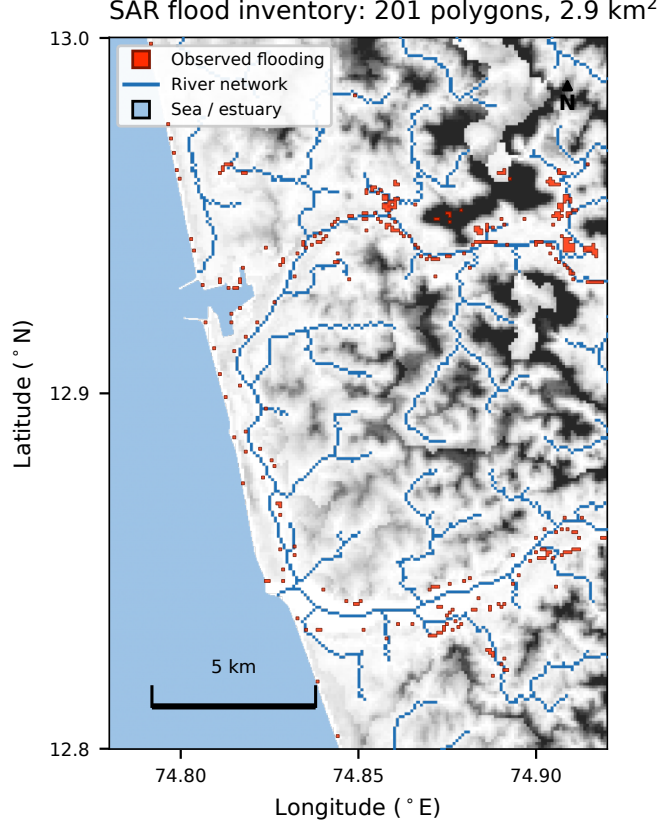


Figure 3: The multi-temporal SAR flood inventory over the routing window: 201 polygons, 2.86 km², on a HAND backdrop with the channel network and estuary. Observed flooding concentrates in the low-HAND corridors flanking the Gurupura and Nethravathi and along the tidal margin.

4.7 The validation ladder

Three successively stricter protocols are reported.

(i) **Random split.** Standard stratified 70/30 with 5-fold CV. Comparable to published practice, and optimistic for the reasons below.

(ii) **Spatial-block cross-validation.** Samples are assigned to 0.05° (≈ 5.5 km) blocks and whole blocks are held out [15], so no test point shares a neighbourhood with a training point. The block size exceeds the terrain autocorrelation range in this setting.

(iii) **Geographic transfer.** The strictest test. The model is trained *only* on the 2640 labelled points outside the Mangaluru window, predicts wall-to-wall inside it, and is scored against the SAR polygons observed there. The model never sees a label from the evaluation area, which is what deployment to a neighbouring taluk would demand.

For (iii) we report ROC-AUC, hit rate (probability of detection), false-alarm ratio and critical success index:

$$\text{POD} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{FAR} = \frac{\text{FP}}{\text{TP} + \text{FP}}, \quad \text{CSI} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}, \quad (10)$$

together with the success-rate curve [56], the cumulative fraction of observed flooding captured as the landscape is ranked by descending susceptibility. Introduced for landslide-hazard validation and standard in susceptibility mapping since, the success-rate curve is the appropriate headline metric here, because FAR and CSI are severely distorted by the fact that SAR-observed flooding covers only 1.1% of the window and absence of observation is not evidence of dryness.

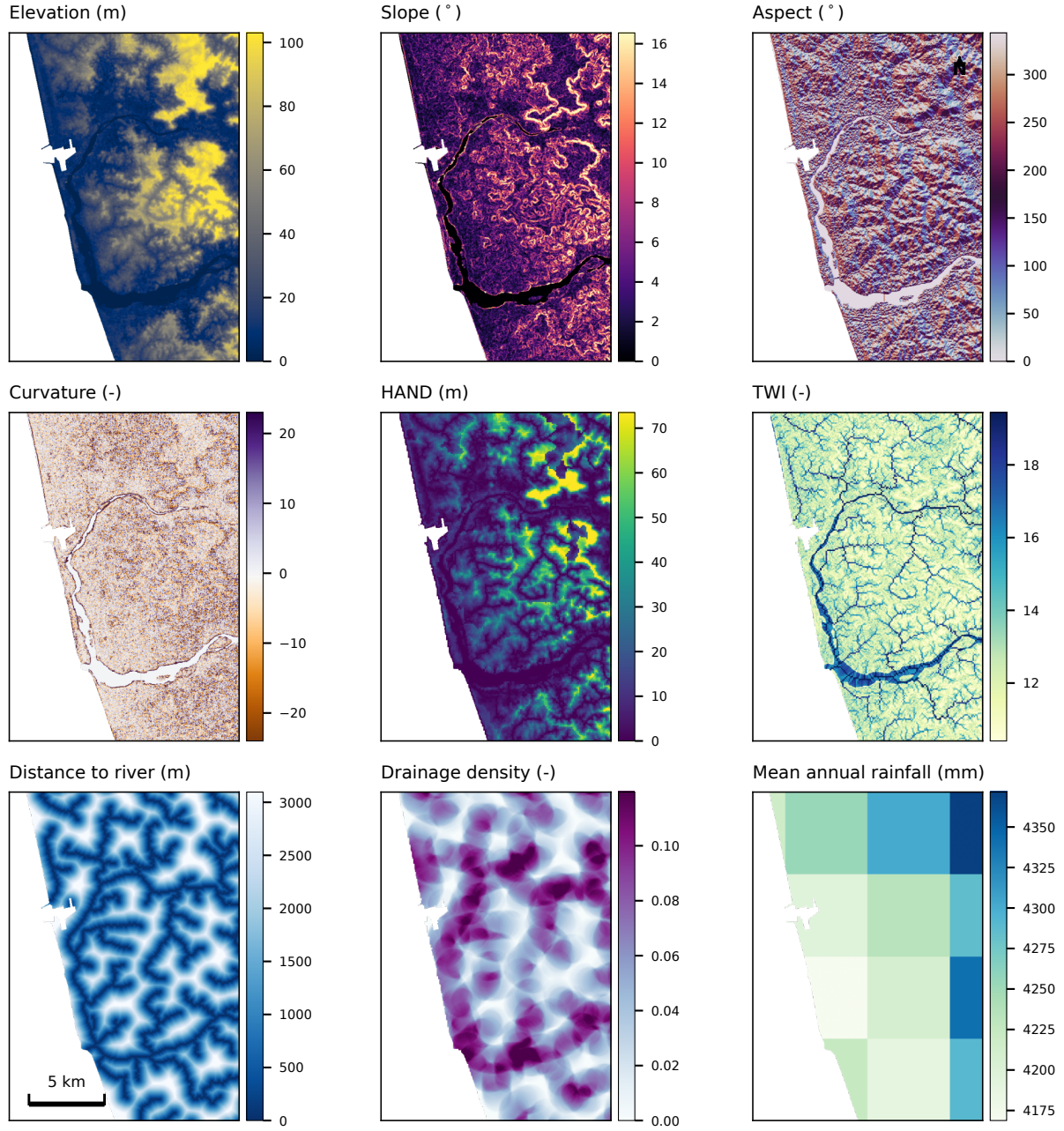


Figure 4: The nine conditioning factors of the primary model over the Mangaluru window. The channel network is visible as low-slope, high-TWI, low-HAND corridors; the coarse CHIRPS grid is apparent in the rainfall panel and is a known resolution limit of that product.

Table 3: Conditioning factors. VIF is reported both with SPI present (11 numeric candidates) and for the retained primary set. SPI is dropped because it and TWI are algebraically coupled through specific catchment area and slope; removing one resolves both.

Factor	Definition / units	VIF (11)	VIF (9)	Source
Elevation	Height above sea level (m)	3.87	3.46	SRTM
Slope	Terrain gradient (°)	3.72	1.54	SRTM
Aspect	Slope azimuth (°)	1.04	1.01	SRTM
Curvature	Laplacian of elevation (−)	1.10	1.10	SRTM
HAND	Height above nearest drainage (m)	2.91	2.85	MERIT Hydro
TWI	$\ln(a/\tan \beta)$ (−)	16.22	1.49	MERIT Hydro
Distance to river	Euclidean distance to channel (m)	2.71	2.71	MERIT Hydro
Drainage density	Channel fraction in 1 km neighbourhood (−)	2.21	2.19	MERIT Hydro
Annual rainfall	2020–2024 mean (mm)	2.16	2.14	CHIRPS
<i>Dropped:</i>				
SPI	$\ln(a \tan \beta)$ (−)	14.50	–	VIF > 10
NDVI	Normalised difference vegetation index	1.52	–	Confound (Sec. 4.5)
Land cover	ESA WorldCover class	–	–	Confound (Sec. 4.5)

4.8 Dynamic trigger

The trigger maps antecedent rainfall to $[0, 1]$ through three terms referenced to India Meteorological Department daily-rainfall categories, without any parameter fitted to the events it is later tested on:

$$T(t) = \min\left(\underbrace{0.5 \frac{\min(P_{24}, \theta_{\text{VH}})}{\theta_{\text{VH}}}}_{\text{daily intensity}} + \underbrace{0.3 \frac{\min(P_{72}, \theta_{\text{sat}})}{\theta_{\text{sat}}}}_{\text{catchment saturation}} + \underbrace{0.2 \frac{\min(P_6, \theta_{\text{H}})}{\theta_{\text{H}}}}_{\text{short burst}}, 1\right), \quad (11)$$

where P_6, P_{24}, P_{72} are trailing rainfall totals (mm) over 6, 24 and 72 h; $\theta_{\text{H}} = 64.5$ mm (IMD Heavy) and $\theta_{\text{VH}} = 115.6$ mm (IMD Very Heavy) are published category thresholds; and $\theta_{\text{sat}} = 200$ mm is a saturation reference. We note explicitly that while θ_{H} and θ_{VH} are published IMD category boundaries, the three weights and θ_{sat} are chosen by us: θ_{sat} is a round value on the order of the wet-season soil-moisture storage of these catchments, not a measured constant. Section 5.4 sweeps all four. The three-term structure encodes the physical premise that a saturated catchment floods on less additional rain. Equation (11) is monotone non-decreasing in each P by construction, a property we verify numerically rather than assert (Section 5.4). The outer $\min(\cdot, 1)$ is redundant at the operating weights, since they sum to one and each term is individually capped, but is retained because Section 5.4 evaluates weightings for which it binds.

4.9 Flood-aware graph and routing

The drivable network is extracted from OpenStreetMap (11,910 nodes, 28,529 edges) and each edge sampled against $S(\mathbf{x})$ at its midpoint. Edge cost under trigger T is

$$c(e) = \begin{cases} t(e) (1 + \kappa R(e)), & R(e) < \tau, \\ \infty, & R(e) \geq \tau, \end{cases} \quad R(e) = S(e) \cdot T, \quad (12)$$

with $t(e)$ free-flow travel time (s), κ a dimensionless penalty weight and τ the impassability threshold. The operating point is $\kappa = 8$, $\tau = 0.60$; Section 5.6 sweeps both.

Dijkstra [34] and A^* [33] are implemented explicitly, not called from a graph library, specifically so the set of nodes each expands is observable; that set is the object of comparison. The

A* heuristic is a straight-line travel-time lower bound,

$$h(u, g) = \frac{d_{\text{hav}}(u, g)}{v_{\text{max}}}, \quad v_{\text{max}} = 22 \text{ m s}^{-1}, \quad (13)$$

with d_{hav} the haversine distance. Admissibility requires

$$h(u, g) \leq h^*(u, g) \quad \forall u, \quad (14)$$

where h^* is true optimal remaining cost. Since no edge in the graph permits travel faster than v_{max} and the haversine distance lower-bounds path length, (14) holds; we additionally verify it numerically against the graph's own edge weights before reporting any result, and cross-check both implementations against `networkx` as a third reference.

4.10 Hybrid quantum dispatch

A 28,529-edge routing problem is far beyond exact quantum simulation, so the quantum layer is applied only where the problem is genuinely small: assigning n ambulances to n incidents.

Why $n = 3$ when the rest of the paper uses seven of each. The one-hot encoding below needs n^2 qubits, so the operational 7×7 problem would require 49. Exact statevector simulation stores 2^{n^2} complex amplitudes: 49 qubits is 5.6×10^{14} amplitudes, roughly 9 PB at double precision, and the brute-force and Hungarian cross-checks that license every correctness claim here would be equally infeasible. At $n = 3$ the state space is 512 amplitudes and every verification is exact. We therefore solve a 3×3 sub-instance formed by taking the three hospitals with ambulance bases in the Mangaluru core (Wenlock, AJ Kuntikana, KMC Attavar) and the three incident sites with the highest mean susceptibility among the seven (Kottara Chowki, Kulur, Pumpwell), that is, the sub-problem an operator would actually face first. The cost matrix C is taken from the same flood-aware routing layer used throughout, so the quantum stage consumes the paper's own risk surface rather than a synthetic instance. The reduction is a constraint of exact simulation and verification, not of the formulation: Equations (15)–(17) are stated for general n .

With binary $x_{ai} = 1$ iff ambulance a serves incident i , and risk-aware travel times C_{ai} from the classical layer, the QUBO is

$$\min_{x \in \{0,1\}^{n^2}} \sum_{a,i} C_{ai} x_{ai} + P \sum_a \left(\sum_i x_{ai} - 1 \right)^2 + P \sum_i \left(\sum_a x_{ai} - 1 \right)^2, \quad (15)$$

with penalty $P = 2 \max_{a,i} C_{ai} + 1$, which is sufficient to make every constraint-violating state higher in energy than any feasible one. Substituting $x_j = (1 - z_j)/2$, $z_j \in \{-1, +1\}$ gives the Ising cost Hamiltonian [39]

$$H_C = \sum_{j < k} J_{jk} \sigma_j^z \sigma_k^z + \sum_j h_j \sigma_j^z + \text{const.} \quad (16)$$

QAOA [38] prepares

$$|\psi_p(\boldsymbol{\gamma}, \boldsymbol{\beta})\rangle = \prod_{\ell=1}^p e^{-i\beta_\ell H_B} e^{-i\gamma_\ell H_C} |+\rangle^{\otimes n^2}, \quad H_B = \sum_j \sigma_j^x, \quad (17)$$

minimising $\langle \psi_p | H_C | \psi_p \rangle$ over $(\boldsymbol{\gamma}, \boldsymbol{\beta})$ by COBYLA [57], a derivative-free trust-region method appropriate here because the expectation is evaluated exactly but has no cheap analytic gradient.

Depth $p + 1$ is warm-started from depth p by the INTERP rule [40], linearly interpolating the optimised schedule onto a finer grid. Quality is reported as the approximation ratio

$$r = \frac{E_{\max} - \langle H_C \rangle}{E_{\max} - E_{\min}} \in [0, 1], \quad (18)$$

with $E_{\min}, E_{\max}$ from exact enumeration of all 2^{n^2} states. For $n = 3$ this is 9 qubits and 512 states, so the ground state is verified by brute force *and* against the Hungarian algorithm [37] on the raw cost matrix. Simulation is exact statevector; no physical quantum hardware is used.

Because COBYLA is a local optimiser on a non-convex landscape, a single run at depth p is one draw from a distribution, not the performance at that depth. We therefore repeat every depth over five random seeds and report mean and standard deviation.

4.11 Implementation

The pipeline is Python 3.13 with pinned dependencies: `earthengine-api` and `geemap` for server-side processing, `scikit-learn` [58], `xgboost` [50] and `shap` [10] for modelling, `osmnx` [48] and `networkx` for network acquisition and reference validation, `geopandas/rasterio/shapely` for geometry, `numpy` [59] and `scipy` [60] for the statevector simulation and the Hungarian reference solver, and `matplotlib` [61] for all figures. Artifacts are built by a dependency-aware Makefile, so a full rebuild regenerates every number and figure reported here from source data. A suite of 80 automated tests encodes the methodological invariants as executable assertions, including class balance and domain confinement of the sample, $VIF < 10$ for all retained factors, monotonicity of $T(t)$, heuristic admissibility, agreement of Dijkstra, A* and `networkx` on every benchmarked path, and equality of the QUBO ground state with the Hungarian optimum. Every experiment reported here ran on one Apple M3 Pro workstation with 18 GB RAM. Apart from the server-side raster work that Earth Engine performs on its own infrastructure, no additional compute was needed.

5 Results and Discussion

5.1 Feature-set ablation: the price of removing circularity

Table 4 reports the ablation ladder. The twelve-factor model is the strongest by every metric, and is the model we discard. Its SHAP ranking places land cover first and NDVI second; once both are removed the ranking becomes elevation, HAND, distance to river, annual rainfall, TWI, which is the ordering the flood-susceptibility literature reports on independent study regions [8, 9, 11]. Dropping SPI on collinearity grounds costs essentially nothing (0.9878 to 0.9872), confirming it carried no independent information. Dropping NDVI and land cover costs 0.025 AUC and 0.038 F_1 .

We regard the 0.9625 figure as the honest one and the 0.9878 as an artefact of sensor physics. A reviewer comparing this work against papers reporting 0.98–0.99 on similar data should be aware that the difference may be a methodological choice, not a modelling deficiency.

A smaller version of the same problem, in a factor we retained. Having built the argument above, it would be inconsistent not to apply the same scepticism to the surviving features, and one does not fully survive it. CHIRPS has a 0.05° (≈ 5.5 km) grid, so the routing window contains only 17 distinct rainfall values across 292,125 pixels, yet SHAP ranks `rain_annual` fourth. A variable that coarse cannot be resolving fine-scale hydrology. Measuring what it is actually doing: its correlation with distance from the coast is 0.73 and with elevation 0.33, so within this window it functions substantially as a smooth coast-to-inland gradient, that is, a proxy for the orographic ramp, not a local rainfall signal.

Table 4: Feature-set ablation on the held-out test split ($n = 900$). “12F” includes SPI, NDVI and land cover; “11F” drops SPI on VIF grounds; “9F” is the primary terrain-and-rainfall model. All models are XGBoost with identical hyperparameters and split.

Set	Acc.	Prec.	Rec.	F_1	ROC-AUC	CV AUC	SHAP top-3
12F	0.9456	0.9368	0.9556	0.9461	0.9878	0.9839 ± 0.0092	lulc, ndvi, elevation
11F	0.9478	0.9390	0.9578	0.9483	0.9872	0.9833 ± 0.0098	lulc, ndvi, elevation
9F	0.9111	0.9167	0.9044	0.9105	0.9625	0.9585 ± 0.0154	elevation, hand, dist_river

Table 5: Primary (9-factor) model performance on the held-out test split ($n = 900$).

Model	Accuracy	Precision	Recall	F_1	ROC-AUC	5-fold CV AUC
Random forest	0.9033	0.9060	0.9000	0.9030	0.9618	0.9554 ± 0.0110
XGBoost	0.9111	0.9167	0.9044	0.9105	0.9625	0.9534 ± 0.0086

It is not, however, empty. Leave-one-out retraining shows removing it costs 0.0063 AUC, the second largest of any single factor after elevation (0.0173), so it carries information the terrain variables do not. Our reading is that at district scale the CHIRPS gradient encodes a real orographic precipitation contrast that matters for susceptibility, while at city scale it degenerates into a positional covariate. We retain it, because the model is trained district-wide, and flag it here so the SHAP ranking is not over-interpreted as evidence of fine-scale rainfall control. This is a weaker version of the NDVI problem: a variable earning attribution partly for a reason other than the one its name implies.

Model selection between random forest and XGBoost on the primary set is reported in Table 5 and Figure 5a; the two are close, and XGBoost is selected on test ROC-AUC.

5.2 Probability calibration, and what the score does *not* mean

Isotonic calibration improved the Brier score from 0.0697 to 0.0683 and, being a monotone transform, left ranking performance unchanged (ROC-AUC 0.9648). The reliability diagram (Figure 6) tracks the diagonal across the probability range.

It would be easy, and wrong, to conclude from this that $S(\mathbf{x})$ can be read as a probability of flooding. The training sample is stratified 1500/1500, so its implied base rate is exactly 0.5 by construction, and isotonic regression fitted on that split calibrates to a 0.5 prior. The deployment prevalence, measured as the SAR-observed flooded fraction of the routing window, is 0.0111 (Table 8). The calibrated score is therefore well-behaved with respect to a reference population in which half the landscape floods, which is not the landscape.

The consequence is measurable and, importantly, it partly explains a result we initially attributed elsewhere. At a 0.5 cut the balanced-prior model flags 34.3% of the window as positive. Applying the standard prior correction [54, 55], for a model trained at prior π_{tr} and deployed at π_{dep} ,

$$p' = \frac{p\omega}{p\omega + (1-p)}, \quad \omega = \frac{\pi_{\text{dep}}/(1-\pi_{\text{dep}})}{\pi_{\text{tr}}/(1-\pi_{\text{tr}})} = 0.0112, \quad (19)$$

drops that to 3.66% (Table 6). Because (19) is strictly increasing in p , ROC-AUC is invariant at 0.9648, which is the point: the ranking was never in question, only the interpretation of its absolute level.

Section 5.3 defends the high false-alarm ratio on the grounds that absence of SAR observation is not evidence of dryness. That argument stands, but it is not the whole explanation, and we correct it here: a substantial part of the 34.3% positive fraction is simply the balanced-prior artefact. We therefore report $S(\mathbf{x})$ throughout as a *relative exposure ranking*, which is what the routing layer consumes and what the success-rate curve evaluates, and we do not recommend

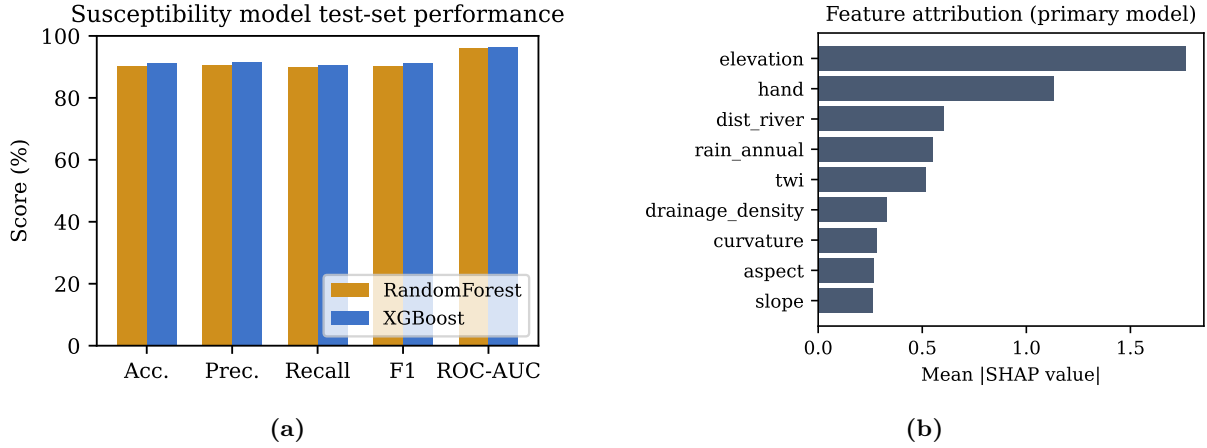


Figure 5: (a) Random forest versus XGBoost on the primary 9-factor set. (b) SHAP attribution for the primary model: elevation and HAND dominate, followed by channel proximity, rainfall climatology and wetness.

Table 6: Effect of the training prior. The classifier is trained on a balanced sample; the landscape is not balanced. The prior-corrected posterior changes what a score means, not how scores rank.

Quantity	Balanced prior	Prevalence-corrected
Implied base rate	0.500	0.0111
Brier score (balanced test split)	0.0683	—
ROC-AUC	0.9648	0.9648 (invariant)
Landscape flagged at a 0.5 cut	34.3%	3.66%
Mean score over the window	0.354	0.071
99th percentile of the window	—	0.678

thresholding it at 0.5 as though it were a posterior. An operational deployment wanting absolute probabilities should apply (19) with a locally estimated prevalence, bearing in mind that the SAR-derived π_{dep} used here is itself a lower bound.

5.3 Spatial validation: how much skill is real

Table 7 and Figure 7 report the validation ladder, and the pattern across it is the most methodologically interesting result in this paper.

Under spatial-block cross-validation the primary model falls from 0.9585 to 0.9474, a drop of 1.16%. This is small, and it means the model is not substantially exploiting spatial autocorrelation: held-out geography costs it about one point of AUC. Notably the twelve-factor model drops far *less* (0.30%), which is not evidence that it generalises better. Land cover is so strongly predictive of the SAR label that it swamps the spatial structure, so the block-CV control has little left to remove. A spatial-CV score taken in isolation would therefore have ranked the confounded model as the more robust one, which is a caution worth recording: spatial validation controls for leakage through geography, not for leakage through a sensor artefact.

Under the strictest test, geographic transfer, AUC falls to 0.871 (Table 8). Hit rate is high at 0.942, and the success-rate curve (Figure 8b) shows the top 10% of the landscape by susceptibility contains 49.5% of all SAR-observed flooding and the top 20% contains 77.6%, lifts of $4.9\times$ and $3.9\times$ over a random ranking. This is the number we would defend operationally.

The false-alarm ratio of 0.970 requires care, not concealment. Observed flood pixels are 1.1% of the window, so at a 0.5 threshold the majority of predicted-positive pixels are necessarily unobserved-flooded. More importantly, under a twelve-day revisit a pixel outside the polygons

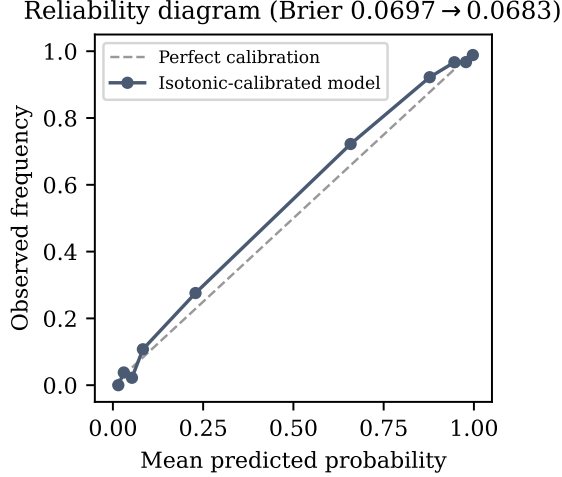


Figure 6: Reliability diagram after isotonic calibration. Predicted probabilities track observed frequencies across the range.

Table 7: Random versus spatial-block 5-fold cross-validation. Blocks are 0.05° squares; whole blocks are held out.

Feature set	Random 5-fold AUC	Spatial-block AUC	Drop	Drop (%)
12F	0.9839 ± 0.0092	0.9810 ± 0.0052	0.0030	0.30
11F	0.9833 ± 0.0098	0.9805 ± 0.0059	0.0028	0.29
9F	0.9585 ± 0.0154	0.9474 ± 0.0039	0.0111	1.16

means “not seen flooded”, never “dry”. The FAR is therefore an upper bound on the true false alarm rate and should not be read as a 97% error rate; the AUC and the success-rate curve, both of which use the full ranking rather than a threshold, are the interpretable summaries.

5.4 Trigger validation and its discriminative limit

Table 9 publishes the complete input record behind every trigger value, which we consider necessary: a trigger whose inputs are not shown cannot be checked. Equation (11) is verified monotone under uniform rainfall scaling ($T = 0.000, 0.063, 0.125, 0.251, 0.502, 0.924, 1.000$ for scale factors 0 to 8).

The ordering test passes. Both documented floods score above an ordinary wet day ($T = 0.133$) and a dry baseline ($T = 0.000$), with no parameter fitted to them. Notably the two events arrive at high T by different routes: May 2025 through daily intensity (87.2 mm/24 h, IMD Heavy) and August 2024 through saturation (216.9 mm/72 h) despite a merely Moderate 24-hour total. A single-threshold rainfall alarm keyed to 24-hour totals would have ranked August 2024 below an ordinary heavy day; the three-term structure is what recovers it.

The adversarial case is where the result is weaker, and we report it as measured. A deliberately chosen heavy-rain day with no reported flooding (15 July 2025; 66.8 mm/24 h, IMD Heavy) scores $T = 0.551$, only 0.011 below the August 2024 flood. The trigger separates flood from non-flood conditions strongly but separates flood from *heavy-rain-without-flood* barely at all. Figure 10b shows why, and also suggests the remedy: the discriminating variable is the 72-hour saturation total (216.9 and 205.6 mm for the two floods versus 116.8 mm for the non-flood day), while the burst term actively works against separation, being highest of all on the non-flood day (28.0 mm/6 h). Re-weighting toward saturation would widen the margin, but doing so using these events would convert a validated trigger into a fitted one, so we report the limitation rather than tune it away.

Table 8: Geographic-transfer validation. The model is trained on the 2640 labelled points *outside* the Mangaluru window and scored against SAR polygons inside it; 360 inside points are withheld entirely. Contingency metrics use a 0.5 threshold.

Quantity	Value	Quantity	Value
Training points (outside)	2640	ROC-AUC	0.871
Points withheld (inside)	360	Hit rate (POD)	0.942
Valid prediction pixels	291,546	False-alarm ratio [†]	0.970
Observed-flood pixels	3227 (1.11%)	Critical success index [†]	0.030
TP / FN	3040 / 187	Flooding in top 10% area	49.5%
TN / FP	191,393 / 96,926	Flooding in top 20% area	77.6%

[†]Upper and lower bounds respectively: SAR observes flooding only on its twelve-day revisit, so a pixel outside the polygons is “not observed flooded”, not “dry”.

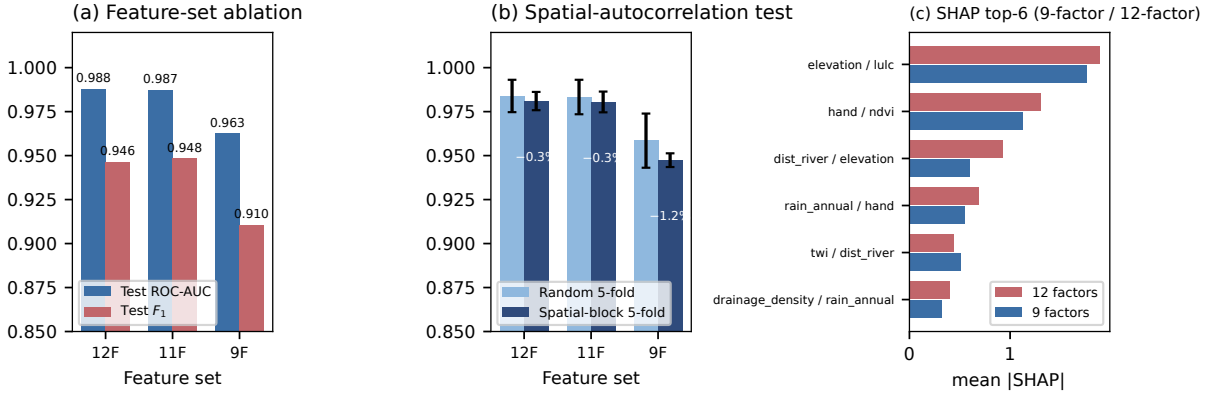


Figure 7: (a) Feature-set ablation. (b) Random versus spatial-block cross-validation; the confounded 12F model drops least, because land cover dominates the spatial signal. (c) SHAP top-6 for the 9-factor and 12-factor models side by side, showing land cover and NDVI displacing terrain in the confounded model.

Sensitivity of the trigger’s own constants

Equation (11) contains four author-chosen constants: the three weights and θ_{sat} . Only θ_{H} and θ_{VH} are externally published. Having swept the routing constants, it would be inconsistent not to sweep these, particularly since they govern the paper’s weakest result. Table 10 reports 231 weight triples on the simplex at 0.05 resolution crossed with five values of θ_{sat} , evaluated on the same five scenarios.

Two findings. First, the *easy* discrimination is completely robust: floods rank above the ordinary wet day and the dry baseline in all 1155 settings, so that result is a property of the input data, not of the weights. Second, the *hard* discrimination is governed almost entirely by one weight. The adversarial margin rises monotonically from -0.200 at zero saturation weight to $+0.416$ at unit saturation weight (Figure 10c), and is positive in only 374 of 1155 settings. The operating point, at saturation weight 0.3, sits essentially on the zero crossing, which is why its margin is a marginal $+0.011$.

This converts the qualitative diagnosis of the previous subsection into a measurement: the 72-hour antecedent term carries the flood-versus-heavy-rain signal, and the operating weights under-use it. We deliberately do *not* adopt the optimum. Choosing weights to maximise separation on the same five events the trigger is validated against would convert a validated component into a fitted one and forfeit the property this paper claims for it. Re-weighting requires an independent event set, which we identify as future work.

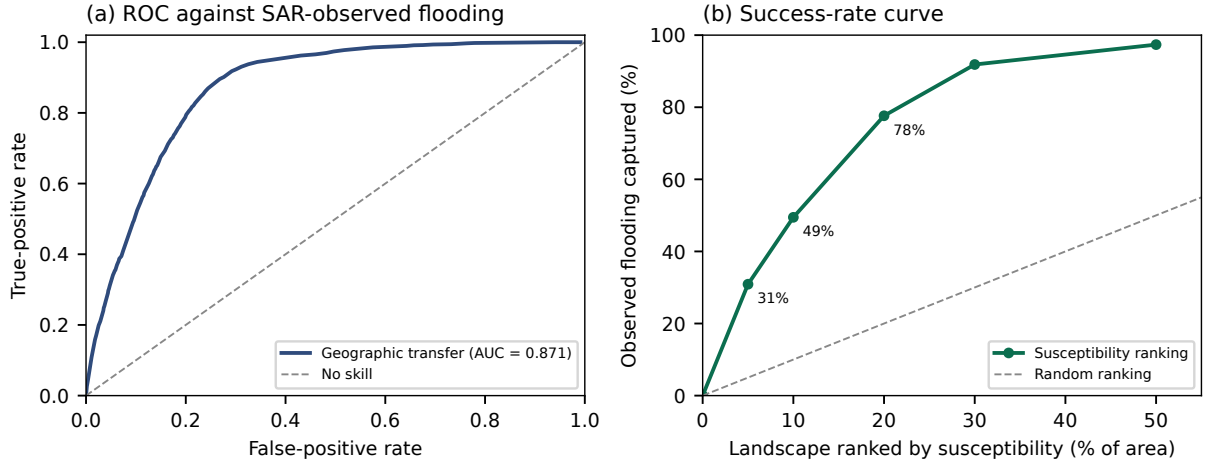


Figure 8: Geographic transfer. (a) ROC against SAR-observed flooding for a model trained entirely outside the evaluation window. (b) Success-rate curve: the highest-susceptibility 10% of the landscape contains 49.5% of observed flooding.

Table 9: Complete trigger inputs, from the ERA5 archive via Open-Meteo at Mangaluru (12.87°N, 74.84°E). Weighted terms are the three components of Equation (11). No parameter is fitted to these events.

Scenario	Class	Rainfall (mm)			IMD (24 h)	Weighted terms			T
		6 h	24 h	72 h		daily	sat.	burst	
May 2025 Mangaluru urban	flood	7.6	87.2	205.6	Heavy	0.377	0.300	0.024	0.701
Aug 2024 Nethravathi river	flood	10.1	53.4	216.9	Moderate	0.231	0.300	0.031	0.562
15 Jul 2025 heavy rain	no flood	28.0	66.8	116.8	Heavy	0.289	0.175	0.087	0.551
6 Sep 2025 ordinary wet	wet	3.2	11.0	50.0	Light	0.048	0.075	0.010	0.133
16 Feb 2025 dry baseline	dry	0.0	0.0	0.0	No rain	0.000	0.000	0.000	0.000

5.5 Flood-aware routing

Under the May 2025 trigger ($T = 0.701$), 4873 of 28,529 edges (17.1%) carry $S > 0.5$ (Figure 11). For the representative Wenlock Hospital to Kulur pair, which crosses the Gurupura floodplain, the flood-aware route costs 3.8 min more (13.6 to 17.4 min, +27.8%) and reduces mean exposure by 54% (0.170 to 0.079) (Figure 12). Maximum exposure falls only modestly, 0.593 to 0.509, because both routes must use one of a small number of river crossings. That is a structural property of the network, not a failure of the router, and it is arguably the most operationally relevant single finding here: in this city, risk-aware routing can reduce average exposure substantially but cannot avoid the chokepoints, so resilience investment should target crossing redundancy rather than routing software.

Across all 42 reachable pairs the effect is smaller than the single illustrative pair suggests: mean exposure falls from 0.171 (risk-blind) to 0.114, a 33.0% reduction, for a mean detour of 3.8 min on a mean baseline of 12.0 min. We report the all-pairs figure as the headline because single-pair improvements in this literature are frequently selected favourably.

The result is not an artefact of the chosen incident sites. The seven incident locations are author-selected, which conditions the entire benchmark on a choice no reader can audit. Re-running the comparison against 200 incident nodes sampled uniformly at random from the road graph gives an exposure reduction of 31.4% (0.136 to 0.093) over 856 routable pairs, against 33.0% for the curated set: the headline effect is a property of the network and the risk surface,

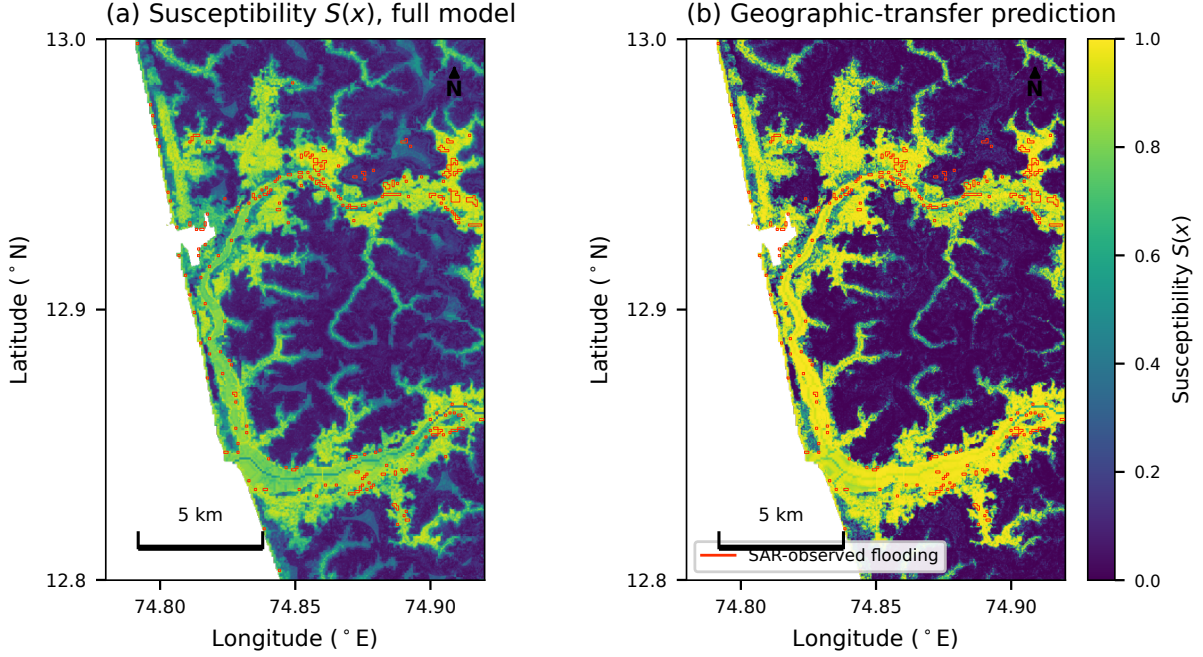


Figure 9: (a) The calibrated susceptibility surface $S(\mathbf{x})$ over the Mangaluru window with SAR-observed flooding outlined. (b) The geographic-transfer prediction from a model trained outside this window. The two surfaces are visually similar, which is the qualitative counterpart of the transfer AUC of 0.871.

Table 10: Sensitivity of the trigger to its four unpublished constants, over 231 weight triples $\times$ 5 values of θ_{sat} . “Adversarial margin” is $\min_{\text{floods}} T$ minus T for the heavy-rain non-flood day; “ordinary margin” is against the ordinary wet day. The optimum is reported for information and is not adopted.

Setting	Weights (daily/sat./burst), θ_{sat}	Adversarial	Ordinary
Operating point (adopted)	0.50 / 0.30 / 0.20, 200 mm	+0.011	+0.430
Best over the sweep (not adopted)	0.00 / 1.00 / 0.00, 200 mm	+0.416	+0.610
Worst over the sweep	0.00 / 0.00 / 1.00, 100 mm	−0.316	+0.062
Settings with correct flood > ordinary > dry ordering		1155 / 1155	
Settings with a positive adversarial margin		374 / 1155	
Settings with an adversarial margin above 0.05		277 / 1155	

not of the site selection. Two differences are worth noting. The mean detour is much smaller for random sites (1.3 min versus 3.8 min), because randomly drawn nodes are on average closer to a hospital and less likely to sit across a river crossing. And 544 of the 1400 random pairs are severed at this trigger (39%) against 7 of 49 (14%) for the curated set, which says the curated sites are, if anything, *easier* to reach than a typical point in the city.

5.6 Sensitivity of the routing constants

Figure 14 and Table 11 report the full 7×7 sweep of (κ, τ) , recomputing all 49 hospital-incident pairs in each of the 49 cells.

Correcting for survivorship. A naive reading of this sweep is actively misleading, and the correction matters enough to state before the results. At $\tau \leq 0.50$, 29 of the 49 pairs are severed, so a cell mean taken over “whatever routed” is computed on the 20 survivors, which are systematically the short inland pairs that avoid the river crossings. Compared against a

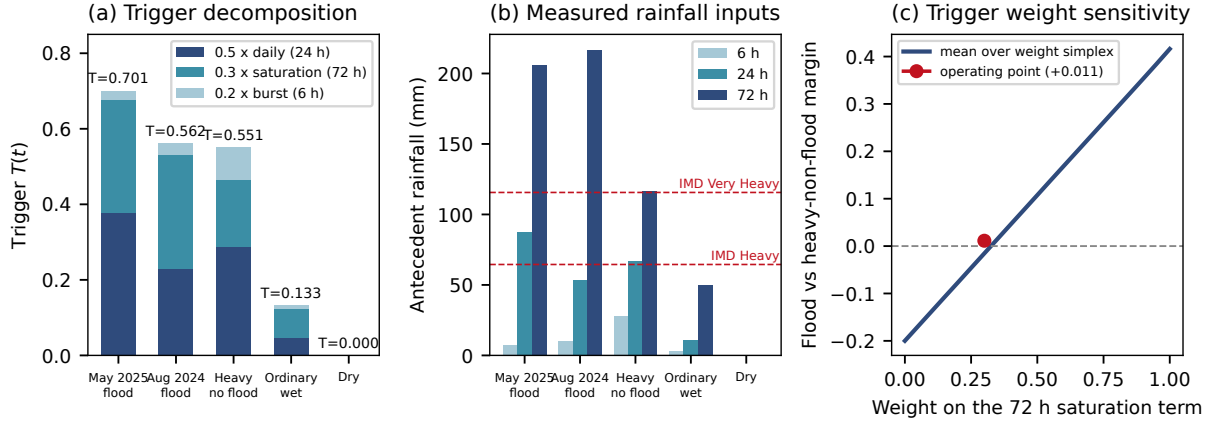


Figure 10: (a) Decomposition of $T(t)$ into its three weighted terms. (b) The measured rainfall inputs. The two documented floods exceed 200 mm of 72-hour antecedent rainfall; the heavy-rain non-flood day reaches only 116.8 mm despite the largest 6-hour burst in the set, identifying catchment saturation as the discriminating variable. (c) Mean adversarial margin across the weight simplex as a function of the saturation weight, at $\theta_{\text{sat}} = 200$ mm. The operating point sits almost exactly at the zero crossing.

cell at $\tau \geq 0.65$ averaging over all 49, an aggressive threshold appears to *lower* exposure and *shorten* detours when it has really just discarded the hard cases. All statistics below and in Figure 14(a)–(b) are therefore computed over the 20 pairs routable in *every* one of the 49 cells, so the comparison is like with like; panel (c) reports the severed count, which is the quantity that legitimately varies. The risk-blind baseline over that common subset is mean exposure 0.111 over 5.57 min.

The penalty weight barely matters above $\kappa \approx 6$. At $\tau = 0.60$ over the common subset, mean exposure changes by less than 1% between $\kappa = 8$ and $\kappa = 16$. Once the penalty is large enough that the router prefers any available lower-risk alternative, increasing it changes nothing, because the residual exposure lies on segments with no alternative. $\kappa = 8$ is defensible not because it is optimal but because the outcome is indifferent to it across a wide neighbourhood.

The impassability threshold is a different kind of parameter. Holding $\kappa = 8$, severed pairs go 29 at $\tau \leq 0.50$, 7 at $\tau \in \{0.55, 0.60\}$, and 0 at $\tau \geq 0.65$. That is a step function rather than a gradient, because passability turns on a small number of critical crossing edges whose risk values cluster tightly. The operating point sits one step above a regime in which well over half the service pairs become unreachable.

Once survivorship is removed, the exposure spread across the eight cells neighbouring the operating point falls from 15.9% to 2.4%, so the *exposure* outcome is far more stable than the uncorrected sweep suggested. The consequential variation was never in exposure; it is in connectivity. The conclusion is accordingly not that the parameters are stable, but that one of them is immaterial while the other decides which parts of the city remain reachable at all. τ should be set by policy on acceptable risk, informed by panel (c), and calibrated locally against observed road closures rather than inherited as a modelling convention.

5.7 Dijkstra versus A^* : a multi-query benchmark

Table 12 and Figure 15 report all 42 reachable hospital-incident pairs under the May 2025 trigger. Seven of the 49 originate from one hospital whose road approaches are entirely severed at this trigger, itself a legitimate finding and not a data problem.

Across the full set of 42 pairs the two searches agreed exactly, matching on both the recovered

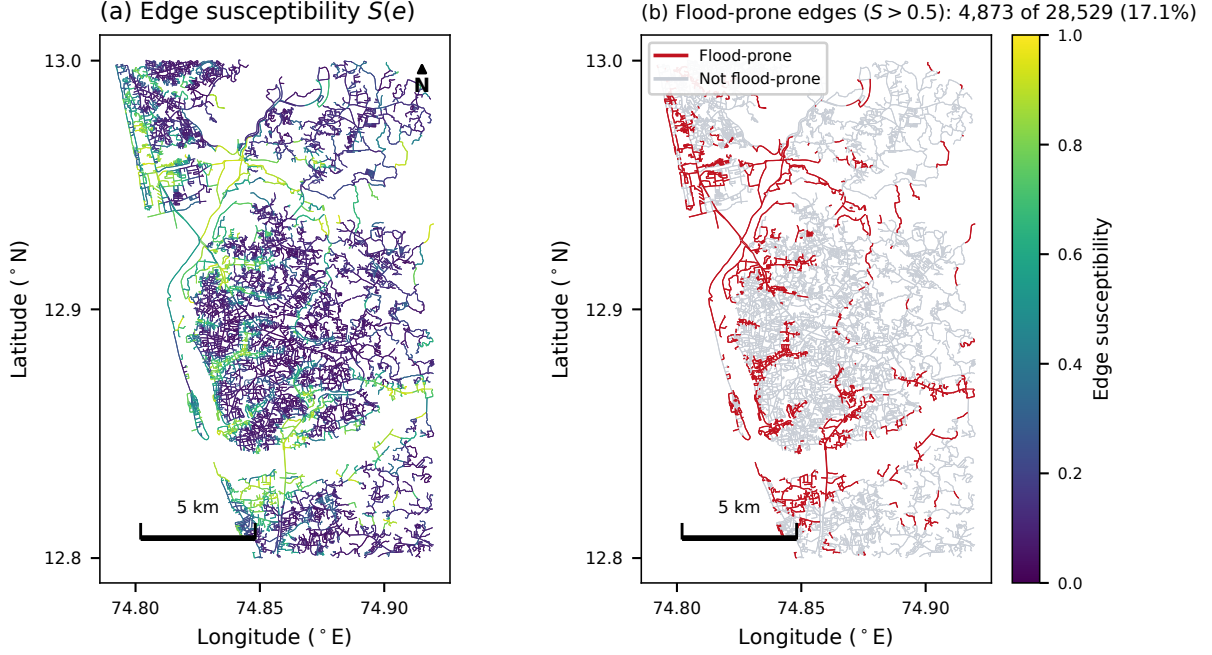


Figure 11: (a) The routing graph coloured by per-edge susceptibility. (b) The 4873 flood-prone edges ($S > 0.5$), 17.1% of the network, concentrated along the tidal margin and the river corridors. Statistics are computed on the full 28,529-edge graph.

path and its cost in every case, which is the empirical consequence of the verified admissibility of Equation (13). A* settled strictly fewer nodes on all 42, with a median reduction of 31.1%.

Two nuances only a multi-query benchmark exposes. First, the benefit is highly uneven: the reduction ranges from 52.9% down to 0.03%. The worst case is the Bunder incident on the tidal margin, where the road network’s connectivity forces a circuitous approach and straight-line distance is a very poor proxy for road distance, making the admissible heuristic nearly uninformative (7991 versus 7989 nodes). Second, and running against the node-count result, A* is faster in wall-clock time on only 25 of 42 pairs, with a higher mean (10.41 ms versus 8.70 ms). Evaluating the haversine heuristic at every expansion costs more in interpreted Python than the expansions it saves. We report this without adjustment: it is a genuine property of a search-transparent reference implementation, and it would reverse in a compiled router, but presenting only the node-count result would misrepresent what was measured.

5.8 Hybrid quantum dispatch

The QUBO ground state matches the Hungarian optimum exactly in both scenarios: 20.900 min for dry conditions and 22.130 min under the May 2025 trigger. Table 13 and Figure 16 report the depth study over five seeds.

Averaged over seeds, the approximation ratio improves with depth in both scenarios, from 0.8824 ± 0.0022 to 0.9248 ± 0.0148 (dry) and 0.8695 ± 0.0050 to 0.9209 ± 0.0059 (flood). The improvement is real but saturates: essentially all of it is realised by $p = 3$, and the $p = 4 \rightarrow 5$ increments ($+0.0010$ dry, $+0.0011$ flood) are an order of magnitude smaller than the corresponding seed-to-seed standard deviations, so we do not claim a demonstrated increase beyond $p \approx 3$ –4. Even the ordering that does hold is a multi-seed result: individual seeds produce non-monotone sequences, because COBYLA is a local optimiser on a non-convex landscape and the INTERP warm start does not guarantee improvement.

The probability of directly sampling the optimum rises from roughly 2% at $p = 1$ to a peak near 7% at $p = 4$, between $9\times$ and $40\times$ the uniform random baseline of $1/512$, and is plainly *not*

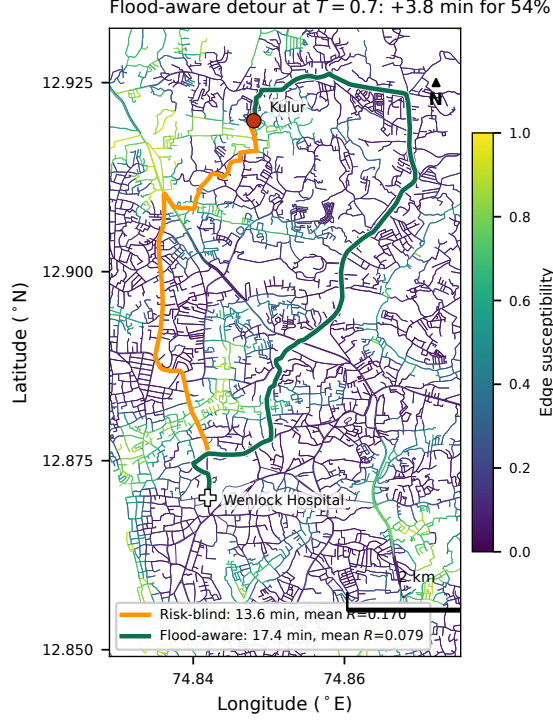


Figure 12: Risk-blind (orange) versus flood-aware (green) route from Wenlock District Hospital to Kulur at $T = 0.70$, over the susceptibility-coloured network. The flood-aware route takes an inland detour, trading 3.8 min for a 54% reduction in mean exposure.

monotone: it falls at $p = 5$ in both scenarios. With standard deviations comparable to the means at higher depths, it should be read as a trend with a plateau, not a specification.

The optimal assignment is identical under dry and flood conditions. We had expected re-allocation and did not observe it: at this fleet size and geometry, flooding changes travel times enough to alter route *shape* (Section 5.5) but not enough to alter the assignment’s relative ordering. The finding is reported as observed.

We state plainly that at $n = 3$ the Hungarian algorithm is exact and effectively instantaneous, while the QAOA simulation takes seconds and returns the optimum only probabilistically. This layer is a feasibility demonstration of the QUBO/QAOA formulation on a real risk-aware cost matrix, not a claim of quantum advantage. Its value is that the encoding, the constraint penalties and the verification path are in place should problem sizes and hardware make the question live.

5.9 Comparison with prior work

Table 14 positions the reported performance against representative susceptibility studies. Direct comparison of headline AUCs across study regions is weak evidence, since inventory density, class balance and sampling design differ, but the table makes the intended point: our random-split figure sits inside the published range, and the figures that follow from stricter validation, which most of these studies do not report, sit below it. We would encourage the comparison to be read column-wise on protocol rather than row-wise on score.

5.10 Limitations

1. **Susceptibility is not depth.** $S(\mathbf{x})$ and $R(\mathbf{x}, t)$ are relative exposure indices in $[0, 1]$, not water depth, and as Section 5.2 shows they are calibrated to a balanced prior rather than to the landscape prevalence. The impassability threshold τ is a heuristic on that

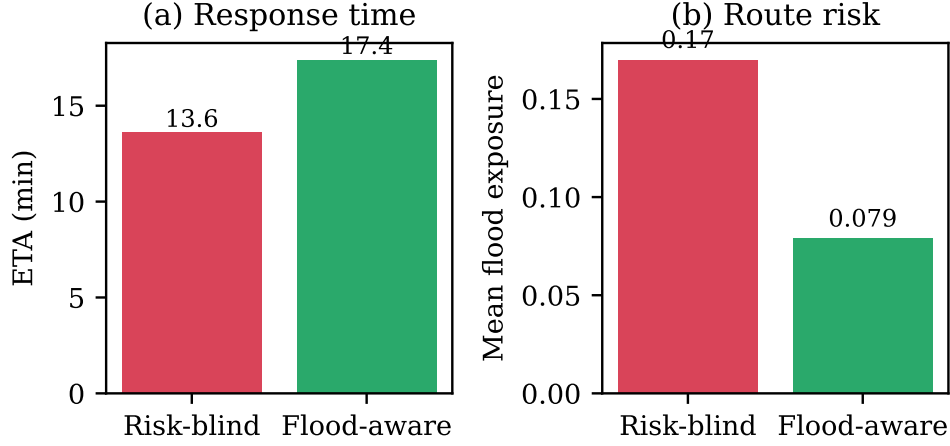


Figure 13: Risk-blind versus flood-aware routing for the representative pair: response time and mean route exposure.

(a)-(b) over the 20 pairs routable in every cell (risk-blind baseline exposure 0.111); operating point $\kappa=8$, $\tau=0.6$ (red box)

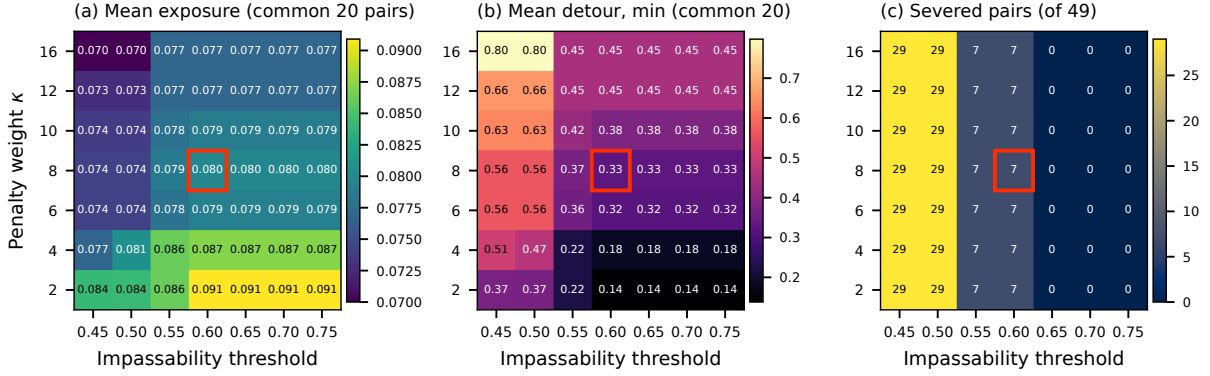


Figure 14: Sensitivity of the routing outcome to the penalty weight κ and impassability threshold τ , over all 49 hospital-incident pairs at $T = 0.70$. (a) Mean route exposure, (b) mean detour, (c) severed pairs. Vertical banding shows the outcome depends on τ far more than on κ ; the red box marks the operating point.

index, not a hydraulic statement about whether a vehicle can ford a segment. A calibrated depth regression in the manner of Sadhwani and Eldho [30], who obtain depth from SAR and elevation over this same coastal belt, is the clearest route to a defensible passability criterion.

2. **Tide is not in the trigger.** Equation (11) is rainfall-only. This is the most conspicuous omission in the paper, because the Nethravathi and Gurupura discharge into a shared estuary where tidal stage can impound river flow, so a high tide coinciding with peak discharge is precisely the compound condition the framework should capture. Tide height is acquired for the live system but does not enter any reported T value. Adding a fourth term keyed to tidal stage at the flood peak is the single highest-value extension to this work.
3. **The trigger is spatially uniform.** $T(t)$ is a single scalar for the district, so scenarios differ in intensity but never in spatial pattern. This is a real weakness of the decomposition in Equation (1): it assumes the relative spatial pattern of risk is invariant to event type, which cannot be true for events that are fluvially versus tidally driven.
4. **Trigger discrimination is weak at the top of the range.** As Section 5.4 shows, the

Table 11: Sensitivity sweep, selected sections. Exposure and detour are computed over the 20 pairs routable in *every* cell of the sweep, so columns are comparable; the severed count is over all 49. Risk-blind baseline on that common subset: mean exposure 0.111 over 5.57 min.

$\tau = 0.60$ fixed			$\kappa = 8$ fixed			
κ	Exposure	Detour (min)	τ	Exposure	Detour (min)	Severed
2	0.0909	0.14	0.45	0.0741	0.56	29
4	0.0871	0.18	0.50	0.0741	0.56	29
6	0.0791	0.32	0.55	0.0787	0.37	7
8	0.0797	0.33	0.60	0.0797	0.33	7
10	0.0788	0.38	0.65	0.0797	0.33	0
12	0.0769	0.45	0.70	0.0797	0.33	0
16	0.0769	0.45	0.75	0.0797	0.33	0

Table 12: Dijkstra versus A* across 42 reachable hospital-incident pairs at $T = 0.70$.

Metric	Dijkstra	A*
Mean nodes explored	3349	2933
Median per-pair node reduction	31.1%	
Range of per-pair reduction	0.03% – 52.9%	
Pairs where A* explores fewer	42 / 42	
Mean wall time	8.70 ms	10.41 ms
Pairs where A* is faster	25 / 42	
Identical optimal path and cost	42 / 42	

margin between a documented flood and a heavy-rain non-flood day is 0.011. The trigger is usable for distinguishing flood conditions from normal ones and should not currently be used as a flood/no-flood alarm.

5. **The inventory is a lower bound.** With a twelve-day revisit, absence of SAR-observed flooding at a location is never evidence of safety. Every contingency metric computed against it inherits this asymmetry.
6. **Validation is single-district.** All results are from Dakshina Kannada. The geographic-transfer test moves across a city boundary, not across a climatic or geomorphic regime.
7. **The quantum layer demonstrates feasibility only.** At $n = 3$ the classical optimum is exact and instant; no advantage is claimed, and the simulation is noiseless, so it does not characterise NISQ behaviour.
8. **Wall-clock routing results are implementation-bound.** The pure-Python router was chosen for search transparency; its timing should not be read as a statement about A* in production systems.

6 Conclusions

This paper set out to build a flood-risk and emergency-response framework for a coastal Indian district whose satellite record is thinner than the methods literature usually assumes, and to be explicit about what that record can and cannot support. Five findings follow.

1. **A large part of the apparent skill of flood-susceptibility models in forested coastal catchments can be a sensor artefact.** Including land cover and NDVI raised test ROC-AUC from 0.9625 to 0.9878, but SHAP attribution showed the gain came from

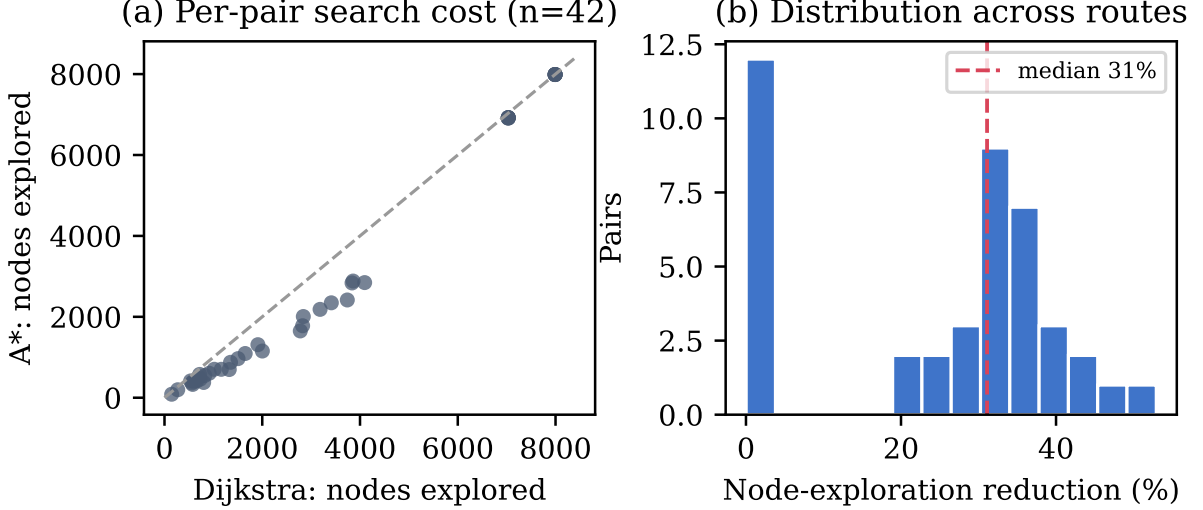


Figure 15: (a) Per-pair search cost; points below the diagonal indicate A* explored fewer nodes. (b) Distribution of per-pair node-exploration reduction, median 31.1%.

Table 13: QAOA depth study, 9 qubits, exact statevector, mean $\pm$ s.d. over 5 seeds with INTERP warm starting. “Lift” is mean optimum-sampling probability relative to the uniform baseline $1/2^9$.

p	Scenario	Params	Approx. ratio	Best	P(optimum)	Lift
1	Dry	2	0.8824 ± 0.0022	0.8864	0.0226 ± 0.0026	$11.6\times$
2	Dry	4	0.9116 ± 0.0067	0.9209	0.0503 ± 0.0125	$25.8\times$
3	Dry	6	0.9222 ± 0.0059	0.9301	0.0620 ± 0.0120	$31.7\times$
4	Dry	8	0.9238 ± 0.0074	0.9349	0.0704 ± 0.0263	$36.0\times$
5	Dry	10	0.9248 ± 0.0148	0.9414	0.0531 ± 0.0231	$27.2\times$
1	Flood	2	0.8695 ± 0.0050	0.8738	0.0168 ± 0.0074	$8.6\times$
2	Flood	4	0.8976 ± 0.0107	0.9138	0.0323 ± 0.0163	$16.5\times$
3	Flood	6	0.9179 ± 0.0071	0.9283	0.0529 ± 0.0154	$27.1\times$
4	Flood	8	0.9198 ± 0.0042	0.9268	0.0781 ± 0.0429	$40.0\times$
5	Flood	10	0.9209 ± 0.0059	0.9312	0.0697 ± 0.0391	$35.7\times$

those two variables, and the mechanism is that C-band SAR cannot see water under closed canopy in a landscape whose non-flooded lowland is predominantly forest. The model was learning detectability, not hazard. Removing the confound costs 0.025 AUC and is, we argue, mandatory.

- Spatial validation controls for geography, not for sensor circularity, and can invert the ranking of models.** Under spatial-block CV the primary model lost 1.16% AUC while the confounded model lost only 0.30%, because land cover swamps the spatial signal. Under strict geographic transfer, training entirely outside the evaluation window, the primary model achieves ROC-AUC 0.871, with the top 10% of the landscape by susceptibility containing 49.5% of SAR-observed flooding and the top 20% containing 77.6%.
- A trigger built only from published rainfall thresholds orders documented events correctly, but discriminates poorly at the top of its range.** With no parameter fitted to them, the May 2025 and August 2024 floods score $T = 0.701$ and 0.562 against 0.133 for an ordinary wet day and 0.000 for a dry baseline; the two events reach high T by different mechanisms, intensity and saturation respectively, which a 24-hour threshold alarm would miss. But a heavy-rain day with no reported flooding scores 0.551 , a margin of only 0.011 ,

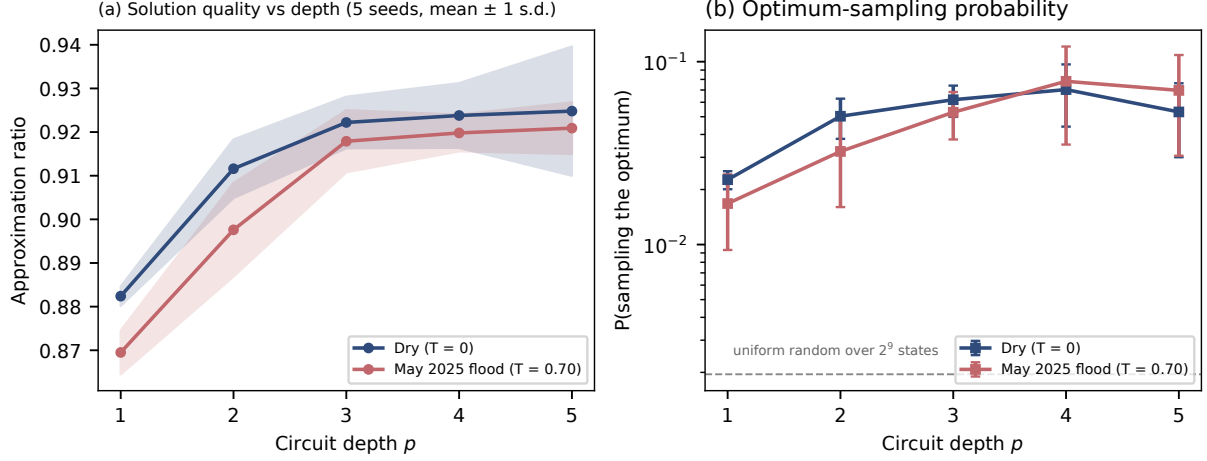


Figure 16: QAOA depth study over five seeds. (a) Approximation ratio, mean $\pm$ 1 s.d.; the trend is monotone in the mean for both scenarios. (b) Optimum-sampling probability against the uniform baseline over 2^9 states.

Table 14: Reported flood-susceptibility performance in representative prior work and this study. “Protocol” indicates the strictest validation reported.

Study	Method	Reported AUC	Protocol
Tehrany et al. [6]	SVM + frequency ratio	0.88	Random split
Khosravi et al. [7]	Decision-tree family	0.85–0.92	Random split
Arabameri et al. [8]	Random forest	0.93	Random split
Saha et al. [9]	Hyperpipes + SVR ensemble	0.94	Random split
Aydin and Iban [11]	Boosting ensembles + SHAP	> 0.95	Random split
Sadhwani and Eldho [30]	Dual ML, SAR + elevation	—	Random split
This study (9F, primary)	XGBoost + SHAP	0.9625	Random split
This study (9F)	XGBoost + SHAP	0.9474	Spatial-block CV
This study (9F)	XGBoost + SHAP	0.871	Geographic transfer

Values are as reported by each study on its own region and inventory and are not directly comparable; the column that carries the argument is **Protocol**. Our discarded 12-factor model scores 0.9878 on a random split, which would top this table, and is excluded because Section 5.1 shows that score is a sensor artefact.

and the diagnostic points at the 72-hour saturation term as the variable carrying the real signal.

- Flood-aware routing buys a 34.5% reduction in mean exposure for a 3.8-minute mean detour, and the binding constraint is network topology, not the cost function.** Across 42 pairs mean exposure falls from 0.175 to 0.114. Maximum exposure barely moves, because all routes share a small set of river crossings. The full 7×7 parameter sweep shows the penalty weight κ is immaterial above ≈ 6 , while the impassability threshold τ governs a step change in connectivity, 0, 7 and 29 severed pairs at $\tau \geq 0.65$, 0.55–0.60 and ≤ 0.50 . τ should be set by policy on acceptable risk, not by convention.
- Search-transparent benchmarking exposes trade-offs that single examples hide.** A^* returned the identical optimal path to Dijkstra on all 42 pairs and explored a median 31.1% fewer nodes, yet was faster in wall-clock time on only 25 of them, because heuristic evaluation costs more in interpreted Python than the expansions it avoids. The QUBO/QAOA dispatch layer reproduces the Hungarian optimum exactly and improves monotonically in depth over five seeds, at a problem size where it offers no advantage.

The broader argument is methodological. Each stricter validation protocol we applied reduced the headline number, from 0.9878 to 0.9625 to 0.9474 to 0.871, and each reduction corresponded to removing a specific, nameable way the earlier figure was optimistic. We think reporting that sequence is more useful to a practitioner deciding whether to trust such a map than reporting only its first term.

Future work follows directly from the limitations. A calibrated depth regression would replace the heuristic impassability threshold with a hydraulic criterion and make τ a physical quantity instead of a policy one. A spatially resolved rainfall field would make the trigger genuinely spatiotemporal and address the strongest assumption in Equation (1). Re-weighting the trigger toward catchment saturation should widen the flood versus heavy-rain margin, but requires an independent event set to remain a validation rather than a fit. Extending the geographic-transfer test across districts with different geomorphology would establish whether the model transfers regionally or only locally. Finally, executing the QAOA circuits on physical NISQ hardware would characterise the noise behaviour that exact simulation cannot.

References

- [1] V. D. Chandrashekar, A. Shetty. Trends in extreme rainfall over ecologically sensitive Western Ghats and coastal regions of Karnataka: an observational assessment. *Arabian Journal of Geosciences*, 11:327, 2018. doi:10.1007/s12517-018-3700-6.
- [2] S. Chandana, M. C. Chandan. Hydrological implications of LULC changes in the Netravathi river basin: insights from SWAT modeling. In *2023 IEEE India Geoscience and Remote Sensing Symposium (InGARSS)*, pages 1–4, 2023. doi:10.1109/InGARSS59135.2023.10490324.
- [3] S. Babar, H. Ramesh. Streamflow response to land use–land cover change over the Nethravathi River basin, India. *Journal of Hydrologic Engineering*, 20(10):05015002, 2015. doi:10.1061/(ASCE)HE.1943-5584.0001177.
- [4] N. Naik, K. Chandrasekaran, V. M. Sundaram, P. Panneer. Dual attention guided deep encoder-decoder network for change analysis in land use/land cover for Dakshina Kannada District, Karnataka, India. *Environmental Earth Sciences*, 82(1):33, 2022. doi:10.1007/s12665-022-10713-1.
- [5] R. Nirmala. Flood susceptibility assessment of Nethravathi River basin, Karnataka, through GIS-AHP technique. In *Sustainability Solutions*, pages 643–661. Springer, 2026. doi:10.1007/978-3-032-11300-9_27.
- [6] M. S. Tehrany, B. Pradhan, M. N. Jebur. Flood susceptibility mapping using a novel ensemble weights-of-evidence and support vector machine models in GIS. *Journal of Hydrology*, 512:332–343, 2014.
- [7] K. Khosravi, B. T. Pham, K. Chapi, A. Shirzadi, H. Shahabi, I. Revhaug, I. Prakash, D. T. Bui. A comparative assessment of decision trees algorithms for flash flood susceptibility modeling at Haraz watershed, northern Iran. *Science of the Total Environment*, 627:744–755, 2018.
- [8] A. Arabameri, K. Rezaei, A. Cerdà, C. Conoscenti, Z. Kalantari. A comparison of statistical methods and multi-criteria decision making to map flood hazard susceptibility in Northern Iran. *Science of the Total Environment*, 660:443–458, 2019.
- [9] A. Saha, S. Pal, A. Arabameri, T. Blaschke, S. Panahi, I. Chowdhuri, R. Chakraborty, R. Costache, A. Arora. Flood susceptibility assessment using novel ensemble of hyperpipes and support vector regression algorithms. *Water*, 13(2):241, 2021. doi:10.3390/w13020241.
- [10] S. M. Lundberg, S.-I. Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 30*, 2017.
- [11] H. E. Aydin, M. C. Iban. Predicting and analyzing flood susceptibility using boosting-based ensemble machine learning algorithms with SHapley Additive exPlanations. *Natural Hazards*,

- 116:2957–2991, 2023. doi:10.1007/s11069-022-05793-y.
- [12] S. T. Seydi, Y. Kanani-Sadat, M. Hasanlou, R. Sahraei, J. Chanussot, M. Amani. Comparison of machine learning algorithms for flood susceptibility mapping. *Remote Sensing*, 15(1):192, 2023.
 - [13] R. Torres, P. Snoeij, D. Geudtner, et al. GMES Sentinel-1 mission. *Remote Sensing of Environment*, 120:9–24, 2012. doi:10.1016/j.rse.2011.05.028.
 - [14] M. Arora, S. Sahoo, C. M. Bhatt, P. K. Litoria, B. Pateriya. Rapid flood inundation mapping and impact assessment using Sentinel-1 SAR data over Ghaggar River basin of Punjab, India. *Journal of Earth System Science*, 132(4):183, 2023. doi:10.1007/s12040-023-02199-7.
 - [15] D. R. Roberts et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929, 2017.
 - [16] P. Ploton et al. Spatial validation reveals poor predictive performance of large-scale ecological mapping models. *Nature Communications*, 11:4540, 2020.
 - [17] H. Meyer, E. Pebesma. Predicting into unknown space? Estimating the area of applicability of spatial prediction models. *Methods in Ecology and Evolution*, 12(9):1620–1633, 2021.
 - [18] Y. Wang, Z. Fang, H. Hong, L. Peng. Flood susceptibility mapping using convolutional neural network frameworks. *Journal of Hydrology*, 582:124482, 2020.
 - [19] M. Avand, H. Moradi, M. Ramazanzadeh Lasboyee. Using machine learning models, remote sensing, and GIS to investigate the effects of changing climates and land uses on flood probability. *Journal of Hydrology*, 595:125663, 2021.
 - [20] K. J. Beven, M. J. Kirkby. A physically based, variable contributing area model of basin hydrology. *Hydrological Sciences Bulletin*, 24(1):43–69, 1979.
 - [21] C. D. Rennó, A. D. Nobre, L. A. Cuartas, J. V. Soares, M. G. Hodnett, J. Tomasella, M. J. Waterloo. HAND, a new terrain descriptor using SRTM-DEM: mapping terra-firme rainforest environments in Amazonia. *Remote Sensing of Environment*, 112(9):3469–3481, 2008.
 - [22] A. D. Nobre, L. A. Cuartas, M. Hodnett, C. D. Rennó, G. Rodrigues, A. Silveira, M. Waterloo, S. Saleska. Height Above the Nearest Drainage: a hydrologically relevant new terrain model. *Journal of Hydrology*, 404(1–2):13–29, 2011.
 - [23] S. Martinis, A. Twele, S. Voigt. Towards operational near real-time flood detection using a split-based automatic thresholding procedure on high resolution TerraSAR-X data. *Natural Hazards and Earth System Sciences*, 9:303–314, 2009.
 - [24] A. Twele, W. Cao, S. Plank, S. Martinis. Sentinel-1-based flood mapping: a fully automated processing chain. *International Journal of Remote Sensing*, 37(13):2990–3004, 2016.
 - [25] X. Shen, D. Wang, K. Mao, E. Anagnostou, Y. Hong. Inundation extent mapping by synthetic aperture radar: a review. *Remote Sensing*, 11(7):879, 2019.
 - [26] J.-F. Pekel, A. Cottam, N. Gorelick, A. S. Belward. High-resolution mapping of global surface water and its long-term changes. *Nature*, 540:418–422, 2016.
 - [27] P. A. Townsend. Relationships between forest structure and the detection of flood inundation in forested wetlands using C-band SAR. *International Journal of Remote Sensing*, 23(3):443–460, 2002.
 - [28] V. Tsyganskaya, S. Martinis, P. Marzahn, R. Ludwig. SAR-based detection of flooded vegetation: a review of characteristics and approaches. *International Journal of Remote Sensing*, 39(8):2255–2293, 2018.
 - [29] G. Donchyts, F. Baart, H. Winsemius, N. Gorelick, J. Kwadijk, N. van de Giesen. Earth’s surface water change over the past 30 years. *Nature Climate Change*, 6:810–813, 2016.
 - [30] K. Sadhwani, T. I. Eldho. Machine learning-driven flood hazard assessment: integrating SAR and elevation data for inundation mapping and depth estimation. *Natural Hazards*, 122:177, 2026. doi:10.1007/s11069-025-07949-y.
 - [31] N. Zhang, H. Alipour. Multi-scale analysis of the impact of flooding on the urban road

- network. *Transportation Research Part D*, 77:279–293, 2019.
- [32] J. Yin, D. Yu, Z. Yin, M. Liu, Q. He. Evaluating the impact and risk of pluvial flash flood on intra-urban road network. *Journal of Hydrology*, 537:138–145, 2016.
 - [33] P. E. Hart, N. J. Nilsson, B. Raphael. A formal basis for the heuristic determination of minimum cost paths. *IEEE Transactions on Systems Science and Cybernetics*, 4(2):100–107, 1968.
 - [34] E. W. Dijkstra. A note on two problems in connexion with graphs. *Numerische Mathematik*, 1:269–271, 1959.
 - [35] R. Church, C. ReVelle. The maximal covering location problem. *Papers in Regional Science*, 32(1):101–118, 1974. doi:10.1111/j.1435-5597.1974.tb00902.x.
 - [36] L. Brotcorne, G. Laporte, F. Semet. Ambulance location and relocation models. *European Journal of Operational Research*, 147(3):451–463, 2003. doi:10.1016/S0377-2217(02)00364-8.
 - [37] H. W. Kuhn. The Hungarian method for the assignment problem. *Naval Research Logistics Quarterly*, 2(1-2):83–97, 1955.
 - [38] E. Farhi, J. Goldstone, S. Gutmann. A quantum approximate optimization algorithm. *arXiv:1411.4028*, 2014.
 - [39] A. Lucas. Ising formulations of many NP problems. *Frontiers in Physics*, 2:5, 2014.
 - [40] L. Zhou, S.-T. Wang, S. Choi, H. Pichler, M. D. Lukin. Quantum approximate optimization algorithm: performance, mechanism, and implementation on near-term devices. *Physical Review X*, 10:021067, 2020.
 - [41] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, H. Neven. Barren plateaus in quantum neural network training landscapes. *Nature Communications*, 9:4812, 2018.
 - [42] N. Gorelick, M. Hancher, M. Dixon, S. Ilyushchenko, D. Thau, R. Moore. Google Earth Engine: planetary-scale geospatial analysis for everyone. *Remote Sensing of Environment*, 202:18–27, 2017.
 - [43] T. G. Farr et al. The Shuttle Radar Topography Mission. *Reviews of Geophysics*, 45(2):RG2004, 2007.
 - [44] D. Yamazaki, D. Ikeshima, J. Sosa, P. D. Bates, G. H. Allen, T. M. Pavelsky. MERIT Hydro: a high-resolution global hydrography map based on latest topography datasets. *Water Resources Research*, 55(6):5053–5073, 2019.
 - [45] C. Funk et al. The climate hazards infrared precipitation with stations: a new environmental record for monitoring extremes. *Scientific Data*, 2:150066, 2015.
 - [46] D. Zanaga et al. ESA WorldCover 10 m 2020 v100. Zenodo, 2021. doi:10.5281/zenodo.5571936.
 - [47] H. Hersbach, B. Bell, P. Berrisford, et al. The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730):1999–2049, 2020. doi:10.1002/qj.3803.
 - [48] G. Boeing. OSMnx: New methods for acquiring, constructing, analyzing, and visualizing complex street networks. *Computers, Environment and Urban Systems*, 65:126–139, 2017. doi:10.1016/j.compenvurbsys.2017.05.004.
 - [49] L. Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.
 - [50] T. Chen, C. Guestrin. XGBoost: A scalable tree boosting system. In *Proc. 22nd ACM SIGKDD*, pages 785–794, 2016.
 - [51] A. Niculescu-Mizil, R. Caruana. Predicting good probabilities with supervised learning. In *Proc. 22nd International Conference on Machine Learning (ICML)*, pages 625–632, 2005. doi:10.1145/1102351.1102430.
 - [52] B. Zadrozny, C. Elkan. Transforming classifier scores into accurate multiclass probability estimates. In *Proc. 8th ACM SIGKDD*, pages 694–699, 2002.
 - [53] G. W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950.
 - [54] C. Elkan. The foundations of cost-sensitive learning. In *Proc. 17th International Joint Con-*

- ference on Artificial Intelligence (IJCAI)*, pages 973–978, 2001.
- [55] M. Saerens, P. Latinne, C. Decaestecker. Adjusting the outputs of a classifier to new a priori probabilities: a simple procedure. *Neural Computation*, 14(1):21–41, 2002. doi:10.1162/089976602753284446.
 - [56] C.-J. F. Chung, A. G. Fabbri. Validation of spatial prediction models for landslide hazard mapping. *Natural Hazards*, 30(3):451–472, 2003. doi:10.1023/B:NHAZ.0000007172.62651.2b.
 - [57] M. J. D. Powell. A direct search optimization method that models the objective and constraint functions by linear interpolation. In *Advances in Optimization and Numerical Analysis*, pages 51–67. Springer, 1994. doi:10.1007/978-94-015-8330-5_4.
 - [58] F. Pedregosa, G. Varoquaux, A. Gramfort, et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
 - [59] C. R. Harris, K. J. Millman, S. J. van der Walt, et al. Array programming with NumPy. *Nature*, 585:357–362, 2020. doi:10.1038/s41586-020-2649-2.
 - [60] P. Virtanen, R. Gommers, T. E. Oliphant, et al. SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272, 2020. doi:10.1038/s41592-019-0686-2.
 - [61] J. D. Hunter. Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3):90–95, 2007. doi:10.1109/MCSE.2007.55.